GLEASON, KOCHEN–SPECKER, AND THE MODEL EXISTENCE GAME

OLIVIA AUBONE AND JOSÉ IOVINO

Abstract. The Kochen–Specker theorem is a statement of two-valued propositional logic: certain finite sets of rays admit no consistent assignment of the values 0 and 1. We show that it is the shadow of a far richer phenomenon in infinitary real-valued logic, where the truth value of a quantum proposition is its probability. We give a countable theory whose standard models are, up to isomorphism, the separable infinite-dimensional complex Hilbert spaces with their lattices of closed subspaces and their quantum measures, so that Gleason's theorem holds in them, and we show that quantum truth values are irreducibly continuous: in the standard models they fill the interval $[0, 1]$ and cannot be confined to any Borel set of values that misses a point of it, and in no model, countable and incomplete ones included, can they be collapsed to 0 and 1. Stating the model existence theorem as a game, we show that every hidden-variable hypothesis expressible as an axiom and incompatible with the theory is refuted by a well-founded tree of finite obstructions; for two-valuedness, we describe a strategy whose leaves are classical Kochen–Specker configurations.

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Introduction

The Kochen–Specker theorem [Bel66, KS67] is a statement of two-valued propositional logic: in a Hilbert space of dimension at least three, certain finite sets of one-dimensional subspaces admit no assignment of the values 0 and 1 in which, for every orthonormal basis among them, exactly one of the subspaces spanned by its vectors receives the value 1. Gleason's theorem [Gle57] concerns probabilities rather than truth values: on a separable Hilbert space of dimension at least three, every σ-additive probability measure on the closed subspaces is given by a density operator, that is, by a positive operator of trace one.

In this paper both theorems pass to the real-valued $L_{\omega_1,\omega}$ of the first author [Aub], in which the truth value of a quantum proposition, a closed subspace, is its probability. We give a countable theory of this logic whose models carry a lattice and a real-valued function on it, the measure; in the standard models, the lattice is isomorphic to that of the closed subspaces of a separable Hilbert space, and the measure is given by a density operator, as in Gleason's theorem. We show that quantum truth values are irreducibly continuous: in the standard models they fill the interval $[0, 1]$ and cannot be confined to any Borel set of values that misses a point of it, and in no model, countable and incomplete ones included, can they be collapsed to 0 and 1. Hidden variables are thus excluded by the logic of quantum propositions itself, in models that need not be complete, and not only in the continuum of Hilbert space. The model existence theorem of this logic, stated as a game, keeps the finite witnesses of the classical theorem: every hidden-variable hypothesis that can be written as an axiom and is incompatible with the theory is refuted by a well-founded tree of finite obstructions.

We call the countable theory described above T. Its vocabulary has a Hilbert sort, a Solèr sort for the lattice, a sequence sort for square-summable sequences of vectors, and a real-valued predicate μ on the Solèr sort for the measure; a model of T is *standard* when its Hilbert sort and its sequence sort are the closed unit balls of Hilbert spaces (see Definition 5.2). For an element a of the Solèr sort of a standard model, we write S_a for the closed subspace spanned by the vectors of the Hilbert sort that belong to a, and P_a for the orthogonal projection onto S_a (see Section 5.2).

Theorem A (Gleason's theorem in real-valued $L_{\omega_1,\omega}$). *There is a countable theory T of real-valued $L_{\omega_1,\omega}$ with the following two properties.*

(1) In every standard model of T, the Hilbert sort is the closed unit ball of a separable infinite-dimensional complex Hilbert space H, the map $a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort onto the lattice $L(H)$ of closed subspaces of H, and there is a unique density operator ρ on H such that $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort.

(2) Conversely, for every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, there is a standard model of T whose Hilbert sort is the closed unit ball of H, whose Solèr sort is $L(H)$, and in which $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in L(H)$.

Theorem B (The range of the measure). *Let T be the theory of Theorem A, and, for a Borel set $B \subseteq \mathbb{R}$, let $\mathbf{1}_B(\mu(a))$ be a formula, given by Lemma 5.1, whose value at an element a of the Solèr sort is 1 when $\mu(a) \in B$ and 0 otherwise. In every standard model of T, the range of μ is the whole interval $[0,1]$. Consequently, if B meets $[0,1]$, then the theory T together with the axiom*

$$\sup_a \mathbf{1}_B(\mu(a)) = 0,$$

which says that no value of μ lies in B, has no standard models.

Theorem C (The Bell–Kochen–Specker theorem in real-valued $L_{\omega_1,\omega}$). *The theory T of Theorem A, together with the axiom $\min(\mu(a), 1-\mu(a)) = 0$, which says that μ takes only the values 0 and 1, has no models, standard or not.*

Theorem D (Finite obstructions). *Let T' be a countable theory of real-valued $L_{\omega_1,\omega}$ that has no models. In the model existence game of Section 10, the player who claims that T' has no models has a winning strategy. For every such strategy, the tree of plays in which that player follows it is well-founded; at each leaf, the commitments made along the play, each of which says that the value of a sentence lies in a closed interval with rational endpoints, form a finite set that is satisfied in no structure. The tree branches infinitely only at the moves that concern countable suprema and infima, and once the other player's answers to those moves are fixed, the plays have bounded length. For T' the theory of Theorem C, every finite subset of T' has a model, and the tree of every winning strategy branches infinitely.*

Classical Kochen–Specker configurations reappear at the leaves of these trees. In Section 10, following the proof of Theorem C, we describe informally a winning strategy for the theory of that theorem; at each leaf of its tree, the final position contains nine orthogonal bases, formed by eighteen vectors, of the space spanned by four vectors of a fixed orthonormal basis (see Figure 12).

Roadmap. The background occupies Sections 1–4. In Sections 5 and 6 we present the logic and the theory T, and in Sections 7–9 we prove Theorems A, B and C. The game and the proof of Theorem D occupy Section 10. Two sections are optional: Section 4 and the remarks of Section 11. In Section A we compare the game with Keisler's forcing.

1. Lattices

Birkhoff and von Neumann [BvN36] observed that the experimentally verifiable propositions about a quantum system correspond to the closed subspaces of a Hilbert space, and that these subspaces form a lattice in which the distributive law fails; the logic of quantum mechanics is therefore not that of a Boolean algebra. In place of the distributive law they proposed the weaker modular law, which holds in finite dimensions but fails for the closed subspaces of an infinite-dimensional Hilbert space. What holds in every dimension is the still weaker orthomodular law. In this section we collect the notions from lattice theory that the paper uses: orthocomplementation and orthomodularity, atoms, and the MacNeille completion, which supplies the joins and meets that an incomplete lattice lacks.

Definition 1.1. A *partially ordered set* is a set with a reflexive, antisymmetric and transitive relation $\leq$. For a subset S of a partially ordered set, an *upper bound* of S is an element b with $a \leq b$ for all $a \in S$, and the *join* $\bigvee S$ of S is its least upper bound, when it exists; lower bounds and the *meet* $\bigwedge S$ are defined dually. A *lattice* is a partially ordered set in which every pair of elements a, b has a join $a \vee b$ and a meet $a \wedge b$. A lattice is *bounded* when it has a least element 0 and a greatest element 1, and *complete* when every subset has a join.

In a lattice, the order can be recovered from the meet: the relation $a \leq b$ holds if and only if $a \wedge b = a$. Completeness, likewise, can be stated through meets: a lattice is complete if and only if every subset has a meet, since the meet of a subset is the join of its lower bounds.

Definition 1.2. An *orthocomplementation* on a bounded lattice Λ is a map $a \mapsto a^\perp$ from Λ to Λ such that, for all $a, b \in \Lambda$, the inequality $a \leq b$ implies $b^\perp \leq a^\perp$, and $a^{\perp\perp} = a$, $a \wedge a^\perp = 0$ and $a \vee a^\perp = 1$. A bounded lattice with an orthocomplementation is an *ortholattice*. Two elements a, b of an ortholattice are *orthogonal*, written $a \perp b$, when $a \leq b^\perp$. An ortholattice is *orthomodular* when, for all a, b, the inequality $a \leq b$ implies $b = a \vee (b \wedge a^\perp)$.

In the lattice $L(H)$ of closed subspaces of a Hilbert space H, ordered by inclusion and orthocomplemented by $M \mapsto M^\perp$, orthomodularity says that every closed subspace N containing a closed subspace M is the orthogonal sum of M and $N \cap M^\perp$. Every ortholattice satisfies the De Morgan law $(\bigvee S)^\perp = \bigwedge\{a^\perp : a \in S\}$ for every subset S whose join exists, since $a \mapsto a^\perp$ is an order-reversing bijection that is its own inverse.

Definition 1.3. An *atom* of a bounded lattice is an element $p \neq 0$ such that $b \leq p$ implies $b = 0$ or $b = p$. A bounded lattice is *atomistic* when every element a is the join of the atoms below it (this join being required to exist). It has the *covering property* when, for every atom p and every element a with $p \not\leq a$, no element lies strictly between a and $a \vee p$. It is *irreducible* when, for any two distinct atoms p and q, some atom r distinct from p and q lies below $p \vee q$.

For complete atomistic lattices with the covering property, irreducibility in this sense is equivalent to the absence of a decomposition as a direct product of two lattices, each with more than one element (see Maeda and Maeda [MM70]); we adopt the definition above because it is the form in which irreducibility enters the coordinatization theorem (see Theorem 3.3).

Definition 1.4. A *sub-ortholattice* of an ortholattice Λ is a subset that contains 0 and 1 and is closed under the operations $\vee$, $\wedge$ and $^\perp$ of Λ. An *embedding* of ortholattices is an injective map that preserves $\vee$, $\wedge$, $^\perp$, 0 and 1.

The MacNeille completion, which we construct below, is characterized by a notion of density that concerns the order alone; no topology is involved. Let Γ be a complete lattice, and let Λ be a subset of Γ. We say that Λ is *join-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigvee\{a \in \Lambda : a \leq g\}$, the join being computed in Γ, and that Λ is *meet-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigwedge\{a \in \Lambda : a \geq g\}$. For instance, $\mathbb{Q}$ is join-dense and meet-dense in the extended real line $[-\infty, \infty]$.

A join-dense and meet-dense subset determines, up to isomorphism, the complete lattice that contains it (see Proposition 1.5 below), and that lattice can therefore be built from the subset alone. The construction is due to MacNeille [Mac37]. For a partially ordered set Λ and a subset X of Λ, let X^u and X^l be the sets of upper and lower bounds of X in Λ, and call X a *cut* when $X = (X^u)^l$. The cuts, ordered by inclusion, form a complete lattice $\mathrm{DM}(\Lambda)$, the *MacNeille completion* of Λ, into which the map $a \mapsto \{b \in \Lambda : b \leq a\}$ embeds Λ, preserving all joins and meets that exist in Λ. The letters DM recall its other name, the Dedekind–MacNeille completion.

Proposition 1.5 (Banaschewski [Ban56], Schmidt [Sch56]). *Let Λ be a partially ordered set, and let Γ be a complete lattice that contains Λ as a join-dense and meet-dense subset, the order of Γ inducing on Λ the order of Λ. For $g \in \Gamma$, put $\Lambda_{\leq g} = \{a \in \Lambda : a \leq g\}$. Then $g \mapsto \Lambda_{\leq g}$ is an isomorphism of Γ onto $\mathrm{DM}(\Lambda)$ that restricts on Λ to the canonical embedding $a \mapsto \{b \in \Lambda : b \leq a\}$. If Λ is an ortholattice, then Γ has at most one orthocomplementation that extends that of Λ, and any such orthocomplementation is given by $g^\perp = \bigwedge\{a^\perp : a \in \Lambda,\ a \leq g\}$.*

Proof. By join-density, we have $g = \bigvee \Lambda_{\leq g}$; hence the upper bounds of $\Lambda_{\leq g}$ in Λ are the elements $b \in \Lambda$ with $b \geq g$. By meet-density, the common lower bounds of these elements are the elements $a \in \Lambda$ with $a \leq g$; thus $\Lambda_{\leq g}$ is a cut. For $g, g' \in \Gamma$, join-density also shows that $g \leq g'$ if and only if $\Lambda_{\leq g} \subseteq \Lambda_{\leq g'}$; hence the map is an order embedding, and on Λ it is the canonical embedding, because the order of Γ induces that of Λ. If X is a cut and $g = \bigvee X$ in Γ, then X is contained in $\Lambda_{\leq g}$; conversely, an element $a \in \Lambda$ with $a \leq g$ lies below every $b \in X^u$, since each such b is an upper bound of X in Γ, and therefore $a \in X = (X^u)^l$. Thus $X = \Lambda_{\leq g}$, and the map is onto $\mathrm{DM}(\Lambda)$. The formula for $g^\perp$ follows from $g = \bigvee \Lambda_{\leq g}$ and the De Morgan law. ☺

⧖ **Historical note.** Letters that von Neumann wrote to Birkhoff in 1935, while the two were preparing their paper of 1936, show why they rejected the lattice of closed subspaces of an infinite-dimensional Hilbert space as the logic of quantum mechanics. Their paper postulates instead an abstract projective geometry, whose lattice of subspaces is modular [BvN36, Réd07]. A year later Husimi observed that the closed subspaces of a Hilbert space satisfy the orthomodular law [Hus37], which was rediscovered several times in the 1950s and 1960s. In his thesis of 1964, Piron took completeness, orthomodularity, atoms and the covering property as axioms for the propositions about a quantum system, and he showed that the irreducible lattices with these properties are represented by hermitian forms over a division ring when their dimension is at least four [Pir64].

2. Hermitian spaces

The coordinatization theorem, which we state in Section 3, turns the atoms of a suitable ortholattice into the one-dimensional subspaces of a vector space over a division ring, and its orthocomplementation into orthogonality with respect to a hermitian form. In this section we collect the facts about such spaces that we need. The most important of them, on

which Sections 3 and 4 depend, is that orthomodularity of the lattice of orthogonally closed subspaces is equivalent to a property of the space itself: every orthogonally closed subspace and its orthogonal complement have the whole space as their sum (see Proposition 2.8).

Definition 2.1. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is invertible; multiplication need not be commutative. An *involution* of a division ring Δ is a map $\alpha \mapsto \alpha^*$ such that $(\alpha+\beta)^* = \alpha^* + \beta^*$, $(\alpha\beta)^* = \beta^*\alpha^*$ and $\alpha^{**} = \alpha$ for all $\alpha, \beta \in \Delta$. Let E be a left vector space over Δ. A *hermitian form* on E is a map $\langle \cdot \mid \cdot \rangle \colon E \times E \to \Delta$ such that $\langle \alpha x + \beta y \mid z \rangle = \alpha\langle x \mid z\rangle + \beta\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \langle x \mid y\rangle^*$ for all $x, y, z \in E$ and $\alpha, \beta \in \Delta$; it is *anisotropic* when $\langle x \mid x\rangle = 0$ implies $x = 0$. A *hermitian space* is a left vector space over a division ring with an involution, together with an anisotropic hermitian form.

It follows from these identities that $\langle x \mid \alpha y\rangle = \langle x \mid y\rangle\alpha^*$ for all $x, y \in E$ and $\alpha \in \Delta$. The bracket is linear in its first argument, as is customary in mathematics; in Dirac notation, which physicists use, it is linear in the second. Nothing in the definition forces the scalars to be real or complex: over $\mathbb{Q}$ with the identity involution, the form $\sum_n x_n y_n$, for finitely supported rational sequences $x = (x_n)_{n\in\mathbb{N}}$ and $y = (y_n)_{n\in\mathbb{N}}$, is anisotropic, since a sum of rational squares vanishes only when every square does.

Definition 2.2. In a hermitian space E, two vectors x and y are *orthogonal*, written $x \perp y$, when $\langle x \mid y\rangle = 0$. For $S \subseteq E$, let $S^\perp$ be the set $\{x \in E : \langle x \mid y\rangle = 0 \text{ for all } y \in S\}$. A subspace M is *orthogonally closed* when $M^{\perp\perp} = M$.

Orthogonality is symmetric, since $\langle y \mid x\rangle = \langle x \mid y\rangle^*$.

Definition 2.3 (The lattice $\mathcal{C}(E)$). For a hermitian space E, we write $\mathcal{C}(E)$ for the set of orthogonally closed subspaces of E, ordered by inclusion.

Every set of the form $S^\perp$ is a subspace, and it is orthogonally closed, since $S^{\perp\perp\perp} = S^\perp$; intersections of orthogonally closed subspaces are orthogonally closed. Hence $\mathcal{C}(E)$ is a complete lattice with least element $\{0\}$ and greatest element E; the meet of a family $(M_j)_{j\in J}$ is $\bigcap_j M_j$, and its join is $\big(\sum_j M_j\big)^{\perp\perp}$, which may be strictly larger than $\sum_j M_j$. For $M \in \mathcal{C}(E)$, anisotropy gives $M \cap M^\perp = \{0\}$, whence $M \vee M^\perp = (M^\perp \cap M)^\perp = E$; thus $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is a complete ortholattice.

Definition 2.4. A hermitian space E is *orthomodular* when $M + M^\perp = E$ for every $M \in \mathcal{C}(E)$, where $M + M^\perp$ is the set of sums $v + y$ with $v \in M$ and $y \in M^\perp$.

The lattice join $M \vee M^\perp$ is always E; orthomodularity asks that the algebraic sum be E as well.

Definition 2.5. A *Hilbert space* is a hermitian space H over the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ or the quaternions $\mathbb{H}$, with the usual involution (the identity, complex conjugation or quaternionic conjugation), such that $\langle x \mid x\rangle > 0$ for $x \neq 0$ and H is complete in the norm $\|x\| = \langle x \mid x\rangle^{1/2}$.

Let H be a Hilbert space. By the projection theorem, every closed subspace M of H satisfies $M + M^\perp = H$, and every subspace M satisfies $M^{\perp\perp} = \overline{M}$. Hence the orthogonally closed subspaces of H are exactly its closed subspaces, which form the lattice $L(H) = \mathcal{C}(H)$; the space H is orthomodular, and so, by Proposition 2.8 below, is the lattice $L(H)$.

Lemma 2.6. *In a hermitian space E, every finite-dimensional subspace F is orthogonally closed and satisfies $F + F^\perp = E$.*

Proof. Let n be the dimension of F. By anisotropy, the Gram–Schmidt process gives F an orthogonal basis $w_1, \dots, w_n$ with $\langle w_i \mid w_i\rangle \neq 0$ for $i = 1, \dots, n$. For $z \in E$, the vector $Pz = \sum_i \langle z \mid w_i\rangle\langle w_i \mid w_i\rangle^{-1} w_i$ lies in F, and $z - Pz$ lies in $F^\perp$; hence $F + F^\perp = E$. If $y \in F^{\perp\perp}$,

the vector $y - Py$ lies in $F^\perp \cap F^{\perp\perp}$ and is therefore orthogonal to itself; by anisotropy it is 0, whence $y \in F$.

Lemma 2.7. *Let E be a hermitian space over Δ. If $M \in \mathcal{C}(E)$ and $x \in E$, then $M + \Delta x \in \mathcal{C}(E)$.*

Proof. If $x \in M$, there is nothing to prove; we therefore assume that $x \notin M$. Put $W = M^\perp$, and let ξ_x be the functional $z \mapsto \langle z \mid x\rangle$ on W; it is nonzero, since $x \notin W^\perp = M$, and we may choose $z_0 \in W$ with $\xi_x(z_0) = 1$. Let y be an element of $(M + \Delta x)^{\perp\perp}$, and let ξ_y be the functional $z \mapsto \langle z \mid y\rangle$ on W. Since $W \cap x^\perp = (M + \Delta x)^\perp$, the functional ξ_y vanishes on the kernel of ξ_x; as $z - \xi_x(z)z_0$ lies in that kernel for every $z \in W$, we obtain $\xi_y(z) = \xi_x(z)\alpha$ with $\alpha = \xi_y(z_0)$. Because $\langle z \mid \alpha^* x\rangle = \langle z \mid x\rangle\alpha$, the vector $y - \alpha^* x$ is orthogonal to W and therefore lies in $W^\perp = M$. Thus $y \in M + \Delta x$.

Proposition 2.8. *Let E be a hermitian space. The lattice $\mathcal{C}(E)$ is orthomodular if and only if E is an orthomodular space.*

Proof. Suppose first that E is orthomodular, and let $M \subseteq M'$ be elements of $\mathcal{C}(E)$. Every vector $z \in M'$ can be written as $v + y$ with $v \in M$ and $y \in M^\perp$; since $y = z - v$ lies in M', we have $M' = M + (M' \cap M^\perp) \subseteq M \vee (M' \wedge M^\perp) \subseteq M'$, which is the orthomodular law. Conversely, suppose that $\mathcal{C}(E)$ is orthomodular, let Δ be the division ring of scalars of E, let M be an element of $\mathcal{C}(E)$, and let x be a vector of $E \setminus M$. By Lemma 2.7, the subspace $M' = M + \Delta x$ belongs to $\mathcal{C}(E)$, and orthomodularity gives $M' = M \vee (M' \cap M^\perp)$. Since $M' \neq M$, the subspace $M' \cap M^\perp$ is nonzero; as it meets M only in 0, while M has codimension one in M', it is a one-dimensional subspace Δy. Then $M \subsetneq M + \Delta y \subseteq M'$, and the codimension count gives $M' = M + \Delta y$; hence $x \in M + M^\perp$.

Historical note. In 1964 Piron published a theorem asserting that every orthomodular complex inner product space is complete, but his proof contained an error [Pir64]. Amemiya and Araki gave a correct proof, and their method also covers real and quaternionic scalars [AA66]. It was then speculated that, among infinite-dimensional hermitian spaces over arbitrary division rings, orthomodularity alone characterizes the Hilbert spaces over $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$. Keller ended the speculation in 1980 with a "non-classical Hilbert space", an infinite-dimensional orthomodular hermitian space over a division ring other than these three [Kel80]. Keller was a student of Herbert Gross, whose papers and book on forms in infinite-dimensional vector spaces shaped the theory; the story is told in Holland's survey [Hol95].

3. Projective geometry, Solèr lattices and pre-Solèr lattices

In this section we explain how an ortholattice with a few combinatorial properties can be represented by orthogonally closed subspaces of a vector space with a hermitian form, and how Solèr's condition, together with completeness, forces the scalars to be real, complex or quaternionic. We assume familiarity with Hilbert spaces, but none with projective geometry.

3.1. **Points and lines.** In a complex Hilbert space H, the atoms of $L(H)$ are the one-dimensional subspaces $\mathbb{C}x$, with $x \neq 0$, called *rays*; the join of two distinct rays is the two-dimensional subspace, or *plane*, that they span. More generally, we call the one-dimensional subspaces of any vector space its *rays*. Which rays lie in which planes is information about the lattice alone (see Figure 1); the theorems below show that, when the dimension is at least four, this information determines the vector space and its division ring of scalars, up to isomorphism (see Theorem 3.2).

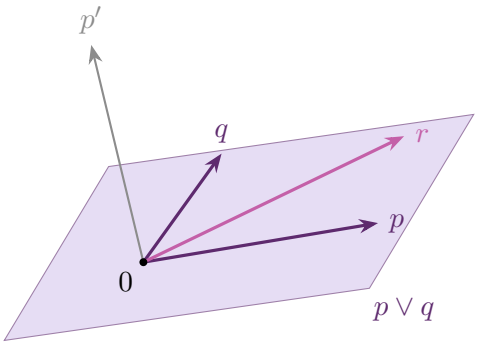


FIGURE 1. Atoms of $L(H)$ are rays. The join $p \vee q$ of two rays is the plane they span; the ray r in that plane is an atom below $p \vee q$, while the ray p' is not.

Definition 3.1. A *projective geometry* is a set of *points* together with a family of subsets, called *lines*, such that:

(PG1) any two distinct points lie on exactly one line;
(PG2) every line contains at least three points;
(PG3) (*the Veblen–Young axiom*) if a, b, a', b' are distinct points and the line through a and b meets the line through a' and b', then the line through a and a' meets the line through b and b'.

A set of points is a *projective subspace* when it contains the whole line through any two of its points; the projective subspace *generated* by a set of points is the intersection of all projective subspaces containing it, and a *projective plane* is the projective subspace generated by three points not on one line. Axiom (PG3) says that two lines lying in a common projective plane always meet (see Figure 2); a projective geometry has no parallel lines. The word *projective* comes from perspective drawing, in which parallel rails, projected onto the canvas, meet at the horizon.

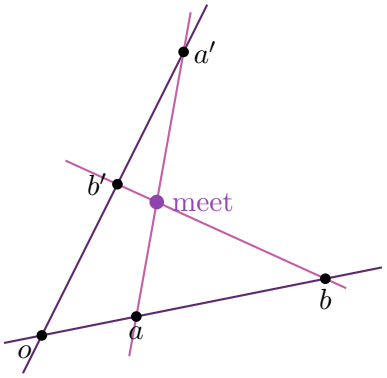


FIGURE 2. The Veblen–Young axiom. The lines ab and $a'b'$ meet at o; then the lines aa' and bb' meet as well.

The basic example comes from linear algebra. Let V be a left vector space of dimension at least two over a division ring Δ. Take as points the rays of V, and as lines the sets of rays contained in a two-dimensional subspace. Axiom (PG1) holds because two distinct rays span a unique two-dimensional subspace, and axiom (PG2) because the two-dimensional subspace

spanned by linearly independent vectors x and y contains the three distinct rays Δx, Δy and $\Delta(x+y)$. For (PG3), suppose that the two-dimensional subspaces spanned by a, b and by a', b' have a ray in common; then the rays a, b, a', b' lie in a subspace of dimension at most three, and, by counting dimensions, any two two-dimensional subspaces of that subspace have a ray in common, in particular those spanned by a, a' and by b, b'.

Points are *independent* when none of them lies in the projective subspace generated by the others; in the basic example, rays are independent exactly when nonzero vectors spanning them are linearly independent. Two projective geometries are *isomorphic* when there is a bijection between their points that maps lines onto lines. A *semilinear isomorphism* from a vector space E over a division ring Δ onto a vector space E' over a division ring Δ' is an additive bijection $f\colon E \to E'$ for which there is an isomorphism $\sigma\colon \Delta \to \Delta'$ with $f(\alpha x) = \sigma(\alpha) f(x)$ for all $\alpha \in \Delta$ and $x \in E$. The following theorem says that, as soon as there are four independent points, the basic example is, up to isomorphism, the only one.

Theorem 3.2 (Coordinatization of projective geometries; see Baer [Bae52]). *Every projective geometry that contains four independent points is isomorphic to the geometry of rays of a left vector space E over a division ring Δ. The pair (Δ, E) is unique up to a semilinear isomorphism.*

Two features of this theorem matter below. First, the division ring Δ is recovered from the geometry alone, and every division ring occurs, among them the rational numbers, the non-archimedean ordered fields (fields with infinitesimals, such as fields of formal Laurent series) and the quaternions. Second, the four independent points cannot be dispensed with: a projective plane, regarded as a projective geometry in its own right, need not come from a division ring. The planes that do are exactly those in which Desargues's theorem holds (if two triangles are in perspective from a point, the three intersection points of corresponding sides lie on one line). Planes in which it fails, the *non-Desarguesian* planes, exist; in a geometry with four independent points, by contrast, Desargues's theorem follows from the axioms.

3.2. **From lattices to geometries.** Let Λ be an atomistic lattice with the covering property. We take its atoms as the points of a geometry whose lines are the sets $\{r : r \text{ an atom with } r \leq p \vee q\}$, for distinct atoms p and q; in $L(H)$, these are the rays in the plane spanned by p and q. The covering property makes this geometry satisfy (PG1) and (PG3), and irreducibility in the sense of Definition 1.3 is exactly (PG2); see Maeda and Maeda [MM70, §§33–34] and Holland [Hol95, §3].

Orthocomplementation adds a relation between points: by the definition in Section 1, atoms p and q are orthogonal when $p \leq q^\perp$. In $L(H)$ this is orthogonality of rays, and it determines the inner product up to the ambiguity described at the end of Theorem 3.3 below. The following properties of an ortholattice Λ enter the theorems of this section:

(C) the lattice is complete;
(OM) it is orthomodular;
(A) it is atomistic;
(CP) it has the covering property;
(I) it is irreducible;
(D) it contains four mutually orthogonal atoms.

Condition (D) supplies the four independent points of Theorem 3.2, since mutually orthogonal atoms are independent. The following theorem, which goes back to Birkhoff and von Neumann [BvN36] in finite dimensions, says that in every ortholattice with (OM), (A), (CP), (I) and (D), orthogonality of atoms is orthogonality of rays with respect to an anisotropic hermitian form, as it is in $L(H)$.

Theorem 3.3 (Coordinatization; Baer [Bae52], Maeda and Maeda [MM70, §34], Holland [Hol95, §3]). *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I) and (D), but not necessarily (C). Then there are a division ring Δ with an involution, a hermitian space E over Δ, and a bijection $p \mapsto R_p$ from the atoms of Λ onto the rays of E, such that two atoms p and q are orthogonal exactly when $R_p \perp R_q$. The map*

$$\gamma\colon \Lambda \to \mathcal{C}(E), \qquad \gamma(a) = \bigcup\{R_p : p \text{ an atom with } p \le a\},$$

is an embedding of ortholattices whose image contains every finite-dimensional subspace of E and the orthogonal complement of each. The triple formed by Δ, E and the form is unique up to a semilinear isomorphism followed by a rescaling $\langle x \mid y\rangle' = \langle x \mid y\rangle\zeta$, where $\zeta^* = \zeta \neq 0$, with the new involution $\alpha \mapsto \zeta^{-1}\alpha^*\zeta$.

Proof of the statements about γ. The orthocomplementation of Λ induces a *polarity* on the geometry of its atoms, in the sense of Holland [Hol95, Definition 3.2]: the map that sends each projective subspace to the set of points orthogonal to all its points. This map reverses inclusion, sends the set of all points to the empty set, and sends each point p to the set of atoms below $p^\perp$, a hyperplane (that is, a maximal proper projective subspace) whose image is again $\{p\}$ (see Holland [Hol95, §4]). By Theorem 3.2, the atoms become the rays of a vector space E, and by Baer's theorem [Bae52] (see Holland [Hol95, Theorem 3.6]), every polarity of this kind under which no point is orthogonal to itself is orthogonality for an anisotropic hermitian form on E. If two atoms lie below a, every atom on the line they determine lies below a; hence $\gamma(a)$ is a subspace. For an atom r, the following are equivalent: $r \le a^\perp$; $a \le r^\perp$; $p \le r^\perp$ for every atom $p \le a$ (because a is the join of these atoms, by (A)); $R_r \perp R_p$ for every atom $p \le a$. Thus $\gamma(a^\perp) = \gamma(a)^\perp$, and applying this twice gives $\gamma(a) = \gamma(a)^{\perp\perp}$. By (A), the relation $a \le b$ holds if and only if $\gamma(a) \subseteq \gamma(b)$; hence γ is injective. Meets are preserved because an atom lies below $a \wedge b$ exactly when it lies below a and b, and joins are preserved by the De Morgan law. Finally, let F be a finite-dimensional subspace of E with basis $x_1, \dots, x_k$, and let p_i be the atom with $R_{p_i} = \Delta x_i$, for $i = 1, \dots, k$. The join $p_1 \vee \cdots \vee p_k$ exists, as every finite join does in a lattice, and, because γ preserves joins, its image is $(\Delta x_1 + \cdots + \Delta x_k)^{\perp\perp} = F^{\perp\perp}$, which is F by Lemma 2.6. The orthogonal complement $F^\perp$ is therefore $\gamma\big((p_1 \vee \cdots \vee p_k)^\perp\big)$. ☺

Piron [Pir64] proved this representation for complete lattices; completeness is not needed for the embedding, and in Theorem 3.6 below it serves only to make γ surjective.

3.3. **Harmonic conjugates and Solèr's condition.** The hypotheses of Theorem 3.3 do not force the division ring Δ to be $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$: the lattice of all subspaces of $\mathbb{Q}^4$, orthocomplemented by orthogonality for the form $\sum_{i<4} x_i y_i$, satisfies them, with $\Delta = \mathbb{Q}$. Solèr's condition, together with completeness, singles out the real numbers, the complex numbers and the quaternions; it is phrased through a classical construction of projective geometry.

Let p and q be distinct points of a projective geometry, write $p + q$ for the line through them (in the basic example, the rays of the plane spanned by p and q), and let c be a point of $p + q$ other than p and q. Two distinct lines in a common projective plane meet in exactly one point, by (PG1) and (PG3), and we denote that point by an intersection sign. Choose a point r not on $p + q$ and a point r' on $r + p$ other than r and p; put $c' = (c + r') \cap (q + r)$ and $c'' = (p + c') \cap (q + r')$. The point $h = (r + c'') \cap (p + q)$ is the *harmonic conjugate* of c with respect to p and q (see Figure 3). In the basic example it does not depend on the choices of r and r', as the computation below shows; by Theorem 3.2, the same holds in every projective geometry with four independent points. In a lattice satisfying (A), (CP) and (I), the same

construction applies to atoms, with $\vee$ in place of $+$ and $\wedge$ in place of $\cap$; the relation "h is the harmonic conjugate of c with respect to p and q" is then expressed by a first-order formula in the language of lattices.

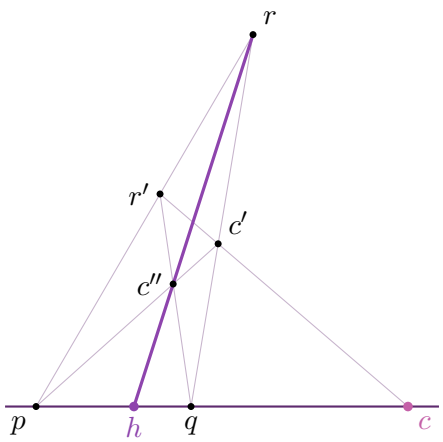


FIGURE 3. The harmonic conjugate h of c with respect to p and q.

In the basic example, the construction is simple in coordinates. Let u and v be vectors spanning p and q, respectively, scaled so that $c = \Delta(u + v)$; this is possible because c lies on $p + q$ and differs from p and q. Likewise, let w be a vector spanning r, scaled so that $r' = \Delta(u + w)$, which is possible because r' lies on $r + p$ and differs from r and p. A direct computation gives $c' = \Delta(v - w)$, $c'' = \Delta(u - v + w)$ and $h = \Delta(u - v)$. All coefficients are ± 1, so the computation is valid over every division ring:

$$\text{the harmonic conjugate of } \Delta(u+v) \text{ with respect to } \Delta u \text{ and } \Delta v \text{ is } \Delta(u-v). \tag{1}$$

Suppose now that the vector space carries an anisotropic hermitian form, as the space E of Theorem 3.3 does, and that $p \perp q$; then $u \perp v$. The orthocomplement of c within the plane $p \vee q$, the ray $c^\perp \wedge (p \vee q)$, consists of the vectors $\alpha u + \beta v$ with $\langle \alpha u + \beta v \mid u + v \rangle = \alpha \langle u \mid u \rangle + \beta \langle v \mid v \rangle = 0$. It coincides with the harmonic conjugate $\Delta(u - v)$ exactly when $\langle u \mid u \rangle = \langle v \mid v \rangle$, that is, when u and v have the same "length" (see Figure 4). In characteristic 2, that is, when $1 + 1 = 0$ in Δ, the vectors $u + v$ and $u - v$ coincide, and the equality $\langle u \mid u \rangle = \langle v \mid v \rangle$ would make $u + v$ isotropic, since then $\langle u + v \mid u + v \rangle = (1 + 1)\langle u \mid u \rangle = 0$. Hence, when the harmonic conjugate of c coincides with its orthocomplement in the plane $p \vee q$, the characteristic of Δ is not 2.

Solèr's condition, in the form we use, asks for an infinite orthogonal sequence of atoms along which the orthocomplement and the harmonic conjugate agree at every step:

(S) There are mutually orthogonal atoms p_n, for $n \in \mathbb{N}$, and atoms $c_n \leq p_n \vee p_{n+1}$ with $c_n \neq p_n, p_{n+1}$, such that, for every $n \in \mathbb{N}$, the harmonic conjugate of c_n with respect to p_n and p_{n+1} is $c_n^\perp \wedge (p_n \vee p_{n+1})$.

This formulation of Solèr's condition through harmonic conjugates is due to Holland [Hol95, Theorems 3.7 and 4.1]; Morash's angle-bisecting axiom [Mor73], which asks for bisectors as in the left half of Figure 4, may be used in its place (see Holland [Hol95, §4]).

Lemma 3.4. *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), and let E be a hermitian space and $p \mapsto R_p$ a bijection from the atoms of Λ onto the rays of*

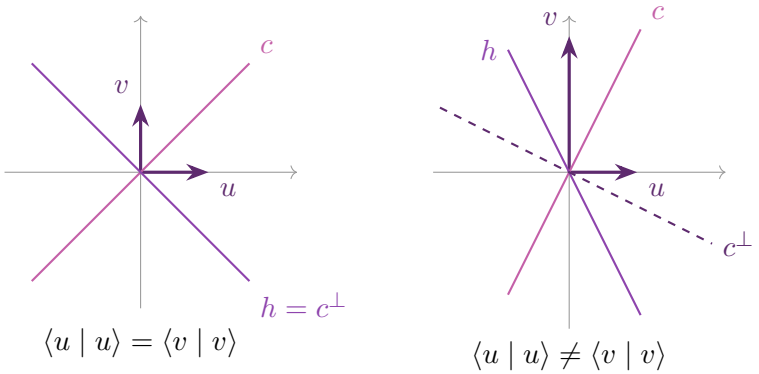


FIGURE 4. The ray $c = \Delta(u+v)$ spanned by the sum of orthogonal vectors u and v, its harmonic conjugate $h = \Delta(u-v)$ with respect to Δu and Δv, and its orthocomplement $c^\perp$ in the plane of u and v. The rays h and $c^\perp$ coincide when u and v have equal length (left), and differ otherwise (right).

E that Theorem 3.3 provides for Λ. After a rescaling of the form, the space E contains an infinite orthonormal sequence.

Proof. Let p_n and c_n, for $n \in \mathbb{N}$, be atoms as in (S). Choose e_0 with $R_{p_0} = \Delta e_0$, and inductively choose e_{n+1} with $R_{p_{n+1}} = \Delta e_{n+1}$ and $R_{c_n} = \Delta(e_n + e_{n+1})$; this is possible because R_{c_n} is a ray of the plane spanned by e_n and e_{n+1}, distinct from Δe_n and Δe_{n+1}. By (1) and the computation that follows it, condition (S) gives $\langle e_n \mid e_n\rangle = \langle e_{n+1} \mid e_{n+1}\rangle$ for every $n \in \mathbb{N}$; call this common value ζ, which satisfies $\zeta^* = \zeta \neq 0$. The form $\langle x \mid y\rangle' = \langle x \mid y\rangle\zeta^{-1}$ is hermitian for the involution $\alpha \mapsto \zeta\alpha^*\zeta^{-1}$, has the same orthogonality relation, and satisfies $\langle e_n \mid e_n\rangle' = 1$ for every n. ☺

Definition 3.5. A *Solèr lattice* is an ortholattice that satisfies (C), (OM), (A), (CP), (I), (D) and (S). A *pre-Solèr lattice* is an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), but not necessarily (C).

Theorem 3.6 (Piron, Amemiya and Araki, Solèr; in this form Holland [Hol95, Theorem 4.1]**).** *An ortholattice is a Solèr lattice if and only if it is isomorphic to $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

Sketch. Let Λ be a Solèr lattice, with γ and E as in Theorem 3.3. Given $M \in \mathcal{C}(E)$, completeness provides the join a of the atoms p with $R_p \subseteq M$, and $\gamma(a)^\perp = \gamma(a^\perp) = M^\perp$, whence $\gamma(a) = M$; thus γ is an isomorphism, $\mathcal{C}(E)$ is orthomodular, and E is an orthomodular space by Proposition 2.8. By Lemma 3.4, the space E contains, after a rescaling, an infinite orthonormal sequence, and Solèr's theorem [Sol95] (see also Prestel [Pre95]) shows that E is a Hilbert space over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Conversely, the lattice $L(H)$ is complete and satisfies the other conditions; for (S), take an orthonormal sequence $(e_n)_{n\in\mathbb{N}}$ in H, and let p_n be the ray of e_n and c_n the ray of $e_n + e_{n+1}$. ☺

Keller [Kel80] constructed orthomodular hermitian spaces that are not Hilbert spaces; hence condition (S) cannot be omitted from Theorem 3.6.

⧗ **Historical note.** Von Staudt's *Geometrie der Lage* of 1847 was the first treatment of projective geometry free of any metric, and his later calculus of "throws" outlined the algebraization of an axiomatic geometry. Hilbert showed in his *Grundlagen der Geometrie* of 1899 that Desargues's theorem follows from the axioms of space but cannot be proved in the plane without

the axioms of congruence, and he founded on it an algebra of segments that does not use those axioms [Hil99]. Veblen and Young gave axioms for projective geometry in 1908 [VY08]; in their book of 1910 they introduced coordinates through an algebra of points that descends from von Staudt's algebra of throws [VY10]. Solèr, a student of Herbert Gross, had just begun work on his orthomodular problem when he died suddenly in 1989; she completed it under the informal guidance of Alexander Prestel, a friend of Gross's at Konstanz, and her theorem appeared in 1995 [Sol95]. In a survey of the same year, Holland restated her condition through harmonic conjugates, constructed as in the book of Veblen and Young, in the form used in this section [Hol95].

4. What a pre-Solèr lattice is missing (optional)

A pre-Solèr lattice satisfies every condition that defines a Solèr lattice except (C). We show that its MacNeille completion supplies the missing elements, and that this completion is a Solèr lattice exactly when it is orthomodular, which need not be the case. Neither completeness nor the orthomodularity of the completion can be expressed, among pre-Solèr lattices, by a sentence of two-valued $L_{\omega_1,\omega}$ about the lattice alone (see Proposition 4.5 below); this is why the vocabulary of Section 5 has vector sorts.

Throughout this section, we fix a pre-Solèr lattice Λ and a hermitian space E that Theorem 3.3 provides for Λ, and we identify Λ with its image $\gamma(\Lambda) \subseteq \mathcal{C}(E)$ under the embedding γ of that theorem.

Theorem 4.1. *Let Λ be a pre-Solèr lattice, identified as above with its image in $\mathcal{C}(E)$. Then Λ is join-dense and meet-dense in $\mathcal{C}(E)$; consequently $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is isomorphic as an ortholattice to the MacNeille completion of Λ.*

Proof. Every $M \in \mathcal{C}(E)$ is the join in $\mathcal{C}(E)$ of the rays it contains, which belong to Λ by Theorem 3.3. If $M^\perp = \{0\}$, then M is E, which belongs to Λ; otherwise M equals $M^{\perp\perp} = \bigcap\{y^\perp : 0 \neq y \in M^\perp\}$, a meet of hyperplanes, each of which belongs to Λ. Both assertions now follow from Proposition 1.5. ☺

The elements of the MacNeille completion that do not come from Λ are therefore orthogonally closed subspaces M such that M and $M^\perp$ are both infinite-dimensional, since Λ contains every finite-dimensional subspace and the orthogonal complement of each (see Theorem 3.3). Adjoining them need not produce a lattice isomorphic to some $L(H)$, because the completion may fail to be orthomodular (see Example 4.3); the next theorem shows that orthomodularity of the completion is exactly what is needed.

Theorem 4.2. *Let Λ be a pre-Solèr lattice, and let E be a hermitian space that Theorem 3.3 provides for Λ. Then the following are equivalent:*

1. *the MacNeille completion of Λ is orthomodular;*
2. *the space E is orthomodular;*
3. *the lattice Λ is isomorphic to a join-dense and meet-dense sub-ortholattice of $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

When these conditions hold, the space E, with its form suitably rescaled, is a Hilbert space as in (3). Moreover, the lattice Λ is isomorphic to $L(H)$ for such an H if and only if Λ is complete.

Proof. Conditions (1) and (2) are equivalent by Theorem 4.1 and Proposition 2.8. If (2) holds, then, by Lemma 3.4 and Solèr's theorem, the space E, with its form rescaled, is a Hilbert space; rescaling changes neither $\mathcal{C}(E)$ nor its orthocomplementation, and (3) follows

by Theorem 4.1. If (3) holds, then, by Proposition 1.5, the lattice $L(H)$, which is orthomodular, is isomorphic, together with its orthocomplementation, to the MacNeille completion of Λ; this is (1). For the last assertion, a complete lattice coincides with its MacNeille completion, and $L(H)$ is complete. ☺

Example 4.3. Let E_0 be the space of finitely supported complex sequences $x = (x_n)_{n\in\mathbb{N}}$, with the inner product $\langle x \mid y\rangle = \sum_n x_n\overline{y_n}$ of the space ℓ^2 of square-summable sequences, and let Λ_0 be the set of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0. Then Λ_0 is a sub-ortholattice of $\mathcal{C}(E_0)$: its elements are orthogonally closed (the finite-dimensional ones by Lemma 2.6), and, for finite-dimensional F and G, the joins $F\vee G = F+G$, $F^\perp \vee G^\perp = (F\cap G)^\perp$ and $F\vee G^\perp = (F^\perp\cap G)^\perp$ belong to Λ_0, and therefore so do the meets, by the De Morgan law. It is orthomodular, because every $M\in\Lambda_0$ satisfies $M + M^\perp = E_0$ by Lemma 2.6, and the argument of Proposition 2.8 applies. Since it is a sub-ortholattice of $\mathcal{C}(E_0)$ that contains every ray of E_0, it inherits (A), (CP) and (I) from $\mathcal{C}(E_0)$, in which (CP) holds by Lemma 2.7; conditions (D) and (S) hold with $p_n = \mathbb{C}e_n$ and $c_n = \mathbb{C}(e_n + e_{n+1})$, where $(e_n)_{n\in\mathbb{N}}$ is the sequence of standard unit vectors. Its MacNeille completion, $\mathcal{C}(E_0)$ by Theorem 4.1, is not orthomodular: by the Amemiya–Araki theorem [AA66], a complex inner product space whose lattice of orthogonally closed subspaces is orthomodular is complete, and E_0 is not. The lattice Λ_0 is thus an orthomodular lattice whose MacNeille completion is not orthomodular; Harding [Har91] studies such lattices.

Orthomodularity of the completion can be stated through the atoms of Λ alone. For a set $\mathcal{A}$ of atoms, let $\mathcal{A}^\perp$ be the set of atoms orthogonal to every element of $\mathcal{A}$, and call $\mathcal{A}$ *orthoclosed* when $\mathcal{A} = \mathcal{A}^{\perp\perp}$. Consider the following condition, named (MOM) for MacNeille orthomodularity:

(MOM) for every orthoclosed set $\mathcal{A}$ of atoms and every atom $p\notin\mathcal{A}$, some atom $q\in\mathcal{A}^\perp$ lies in $(\mathcal{A}\cup\{p\})^{\perp\perp}$.

Theorem 4.4. *The MacNeille completion of a pre-Solèr lattice is orthomodular if and only if the lattice satisfies (MOM).*

Proof. The map $\mathcal{A}\mapsto\bigcup\{R_p : p\in\mathcal{A}\}$, where R_p is the ray of E that corresponds to the atom p under Theorem 3.3, is a bijection from the orthoclosed sets of atoms onto $\mathcal{C}(E)$, carrying $\mathcal{A}^\perp$ to the orthogonal complement of the image of $\mathcal{A}$. If M is the image of $\mathcal{A}$ and $R_p = \Delta x$, then by Lemma 2.7 the set $(\mathcal{A}\cup\{p\})^{\perp\perp}$ corresponds to $M+\Delta x$. Condition (MOM) therefore says that $(M+\Delta x)\cap M^\perp \neq \{0\}$ whenever $M\in\mathcal{C}(E)$ and $x\notin M$. If $y = v+\alpha x$ is nonzero, with $v\in M$ and $y\in M^\perp$, then the scalar α is nonzero, since $y\notin M$, and $x = \alpha^{-1}(y-v)$ lies in $M+M^\perp$; conversely, if $x = v+y$ with $v\in M$ and $y\in M^\perp$, then $y = x-v$ is a nonzero element of $(M+\Delta x)\cap M^\perp$. Thus (MOM) is equivalent to the orthomodularity of E, and Theorem 4.2 applies. ☺

When $\mathcal{A}$ or $\mathcal{A}^\perp$ corresponds to a finite-dimensional subspace, the requirement in (MOM) holds automatically, by Lemma 2.6; the content of (MOM) lies in the orthoclosed sets $\mathcal{A}$ for which both $\mathcal{A}$ and $\mathcal{A}^\perp$ are infinite.

Proposition 4.5. *Neither (C) nor (MOM) is equivalent, among pre-Solèr lattices, to a sentence of two-valued $L_{\omega_1,\omega}$ in the vocabulary of ortholattices expanded by constants for atoms p_n and c_n, for $n\in\mathbb{N}$, that witness (S).*

Proof. Suppose that a sentence φ of $L_{\omega_1,\omega}$ were equivalent to (MOM) among pre-Solèr lattices. The axioms of ortholattices and the conditions (OM), (A), (CP), (I), (D) and (S) are expressed by countably many sentences of $L_{\omega_1,\omega}$ in this vocabulary; together with φ,

they hold in $L(\ell^2)$. By the downward Löwenheim–Skolem theorem for a countable fragment, they hold in a countable structure Λ', which is a pre-Solèr lattice satisfying (MOM). By Theorems 4.4 and 4.2, the lattice Λ' contains every ray of an infinite-dimensional Hilbert space, and is therefore uncountable, a contradiction. The same argument applies to (C), using Theorem 3.6. ☻

⧖ **Historical note.** MacNeille introduced the completion by cuts in 1937 [Mac37], generalizing the cuts by which Dedekind had constructed the real numbers from the rationals in 1872 [Ded72]. Banaschewski and Schmidt characterized it in 1956 by join-density and meet-density [Ban56, Sch56]. In 1969 Adams solved negatively Problem 36 of the third edition of Birkhoff's *Lattice Theory* by giving an orthocomplemented modular lattice whose completion by cuts is not orthomodular [Ada69]. His lattice consists of the finite-dimensional subspaces of an incomplete complex inner product space and their orthogonal complements, as in Example 4.3, and his argument also rests on the Amemiya–Araki theorem. Harding gave in 1991 a construction of orthomodular lattices whose completions by cuts are not orthomodular that avoids that theorem: following Kalmbach, it glues together the Boolean algebras generated by the chains of a lattice [Har91]. Harding noted that whether every orthomodular lattice can be embedded in a complete one was still open.

5. The logic and the vocabulary

Completeness, the one condition in the definition of a Solèr lattice that a pre-Solèr lattice need not satisfy, cannot be expressed by sentences of $L_{\omega_1,\omega}$, two-valued or real-valued, about the lattice alone. The obstacle is the downward Löwenheim–Skolem theorem. For every countable fragment, a Solèr lattice has a countable elementary substructure that contains atoms witnessing (S); this substructure is a pre-Solèr lattice, but it is not a Solèr lattice, since every Solèr lattice is isomorphic to the lattice of closed subspaces of an infinite-dimensional Hilbert space and is therefore uncountable (for the details in the two-valued case, see the optional Proposition 4.5). Axioms about the lattice alone therefore cannot demand the joins that a pre-Solèr lattice may lack, but a Hilbert space placed beside the lattice can supply them. In this section we recall the logic and present a vocabulary that places the lattice beside a Hilbert space and a space of sequences of vectors, and that connects the three by function symbols for orthogonal projection, for the passage between vectors and atoms, for the entries of a sequence, and for its closed span.

5.1. **Real-valued $L_{\omega_1,\omega}$.** We use the real-valued infinitary logic of the first author [Aub], which extends both Eagle's infinitary $[0,1]$-valued logic [Eag14] and Keisler's real-valued logic [Kei23]; the model existence theorem for two-valued $L_{\omega_1,\omega}$ (see Keisler [Kei71]) extends to it. Of this logic we use only the following features:

- ♠ The vocabulary is countable and many-sorted; no metric is given a priori, and function symbols need not be continuous.
- ♠ Predicate symbols are real-valued, each with values in a fixed bounded interval; every formula therefore takes its values in a bounded interval, whose bound depends on the formula.
- ♠ Formulas are closed under continuous connectives, under the quantifiers sup and inf over each sort, and under countable suprema $\sup_n \varphi_n$ and infima $\inf_n \varphi_n$ of families $(\varphi_n)_{n\in\mathbb{N}}$ of formulas that are uniformly bounded and have finitely many free variables in all.

- ♠ There is no equality symbol; as in Keisler's logic [Kei23], a structure is *reduced* when any two elements that agree in every atomic formula with parameters are equal. Models are taken to be reduced.

A *condition* is an expression $\varphi = 0$ or $\varphi \leq 0$, where φ is a formula; an axiom is a condition required to hold for all values of its free variables. Implications between conditions are expressed by the following device: if φ takes only nonnegative values in every structure and r_ψ is a positive rational number with $|\psi| \leq r_\psi$, then the condition

$$\psi \leq \sup_{n \geq 1} \min(r_\psi, n\varphi) \tag{2}$$

says that $\varphi = 0$ implies $\psi \leq 0$, since the right side is 0 when $\varphi = 0$ and r_ψ otherwise. The family $\min(r_\psi, n\varphi)$, for integers $n \geq 1$, is uniformly bounded and has finitely many free variables in all, as a countable supremum requires.

5.2. **The vocabulary.** The vocabulary $\mathcal{L}$ has three sorts. Two of them, the Hilbert sort and the sequence sort, are *vector sorts*, intended to be the closed unit balls of Hilbert spaces; working in unit balls keeps every predicate bounded.

- ♠ The *Hilbert sort* $\mathbf{H}$ has, for each pair λ, λ' of elements of the field $\mathbb{Q}(i)$ of complex numbers with rational real and imaginary parts such that $|\lambda| + |\lambda'| \leq 1$, a binary function $(x, y) \mapsto \lambda x + \lambda' y$; the predicates $\mathrm{Re}\langle x \mid y\rangle$ and $\mathrm{Im}\langle x \mid y\rangle$, with values in $[-1, 1]$, and the predicate $\|x\|$, with values in $[0, 1]$; and constants e_n, for $n \in \mathbb{N}$. We write 0 for the closed term $0e_0 + 0e_0$, and $\langle x \mid y\rangle$ for the pair of predicates $\mathrm{Re}\langle x \mid y\rangle$ and $\mathrm{Im}\langle x \mid y\rangle$.
- ♠ The *sequence sort* $\mathbf{K}$ has the same functions and predicates as $\mathbf{H}$, and one constant ε; in this sort, we write 0 for the closed term $0\varepsilon + 0\varepsilon$.
- ♠ The *Solèr sort* $\mathbf{L}$ has the constants 0 and 1, the binary functions $\wedge$ and $\vee$, the unary function $^\perp$, and a predicate $d(a, b)$ with values in $[0, 1]$, the *discrete predicate*.
- ♠ Function symbols connecting the sorts:

$$\pi \colon \mathbf{L} \times \mathbf{H} \to \mathbf{H}, \quad \mathrm{at} \colon \mathbf{H} \to \mathbf{L}, \quad \mathrm{rep} \colon \mathbf{L} \to \mathbf{H},$$
$$\iota_n \colon \mathbf{H} \to \mathbf{K}, \quad \kappa_n \colon \mathbf{K} \to \mathbf{H} \ (n \in \mathbb{N}), \quad \mathrm{span} \colon \mathbf{K} \to \mathbf{L}.$$

- ♠ A predicate μ on $\mathbf{L}$, the *measure*, with values in $[0, 2]$; the interval is wider than $[0, 1]$ so that the axiom (M1) of Section 6 below, and not the vocabulary, confines μ to $[0, 1]$.

For vector terms t_1, t_2 of the same sort, the equation $t_1 \doteq t_2$ abbreviates the condition $\|\frac{1}{2}t_1 - \frac{1}{2}t_2\| = 0$; for terms of the Solèr sort, the equation $t_1 = t_2$ abbreviates $d(t_1, t_2) = 0$. A vector x *belongs* to an element a of the Solèr sort, or is a *member* of a, when $\pi(a, x) \doteq x$.

The intended meanings are as follows. Each element a of the Solèr sort names a closed subspace of the Hilbert space whose closed unit ball is the Hilbert sort, and the function $\pi(a, \cdot)$ is the orthogonal projection onto that subspace; the members of a are therefore the vectors of the unit ball that lie in it. The predicate d is the discrete metric, with the value 1 at any two distinct elements. Vectors and elements of the Solèr sort are linked by at, which sends a nonzero vector x to the element whose members are the multiples of x in the unit ball, and by rep, which sends a nonzero element a to a unit vector that belongs to a. The constants e_n, for $n \in \mathbb{N}$, form an orthonormal basis. The elements of the sequence sort are square-summable sequences of vectors, linked to the Hilbert sort by ι_n, which places a vector in position n of a sequence, and by κ_n, which reads the entry in position n. The constant ε is

the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$, and the function span sends a sequence s to the element whose members are the vectors of the unit ball in the closed span of the entries of s.

The axiom of Theorem B, which excludes the points of a Borel set from the range of the measure, needs a formula that detects whether a value of the measure lies in that set. For a Borel set $B \subseteq \mathbb{R}$, we write $\mathbf{1}_B$ for its indicator function, which is 1 on B and 0 off B, and we also write $\mathbf{1}_B(\mu(a))$ for any formula of real-valued $L_{\omega_1,\omega}$, with the single free variable a of the Solèr sort, whose value at every element a of every structure for $\mathcal{L}$ is the number $\mathbf{1}_B(\mu(a))$.

Lemma 5.1. *For every Borel set $B \subseteq \mathbb{R}$, there is a formula $\mathbf{1}_B(\mu(a))$ as just described.*

Proof. Let $\mathcal{B}$ be the class of sets $B \subseteq \mathbb{R}$ for which such a formula exists. For rational numbers $\alpha \le \beta$, the distance from a real number t to the interval $[\alpha, \beta]$ is $\max(0, \alpha - t, t - \beta)$, a continuous function of t; hence

$$\inf_{n\ge 1} \max\bigl(0,\, 1 - n\max(0, \alpha - \mu(a), \mu(a) - \beta)\bigr)$$

is a formula, and its value is 1 when $\mu(a) \in [\alpha, \beta]$ and 0 otherwise. Thus $\mathcal{B}$ contains the closed intervals with rational endpoints. If $B \in \mathcal{B}$, then $1 - \mathbf{1}_B(\mu(a))$ is a formula for $\mathbb{R} \setminus B$; if $B_n \in \mathcal{B}$ for every $n \in \mathbb{N}$, then the family of formulas $\mathbf{1}_{B_n}(\mu(a))$ is bounded by 1, and its supremum is a formula for $\bigcup_n B_n$. Hence $\mathcal{B}$ is a σ-algebra. Every open subset of $\mathbb{R}$ is a countable union of closed intervals with rational endpoints, so $\mathcal{B}$ contains every Borel set. ☺

The countable suprema and infima in the formula $\mathbf{1}_B(\mu(a))$ that the proof constructs are nested as in the construction of B from closed intervals. The depth of the nesting is a countable ordinal, and it is unbounded below ω_1 as B varies.

Definition 5.2. A structure $\mathcal{M}$ for $\mathcal{L}$, or for a vocabulary obtained from $\mathcal{L}$ by removing the predicate μ, the sequence sort together with the symbols that involve it, or both, is *standard* when its Hilbert sort is the closed unit ball of a complex Hilbert space H, its sequence sort, if it has one, is the closed unit ball of a complex Hilbert space K, and the functions $\lambda x + \lambda' y$ and the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$ are the restrictions of the operations, the inner products and the norms of these spaces.

A model of one of the theories T_0, T_1 and T of Section 6 below is called *nonstandard* when it is not standard. The word has nothing to do with the nonstandard models of arithmetic or of nonstandard analysis: every predicate takes real values, and no element is infinitely large or infinitesimal. In Lemma 6.3 below, we show that the Hilbert sort of every model of T_0, and hence of every model of T_1 or of T, can be identified with a subset of the closed unit ball of a complex Hilbert space whose $\mathbb{Q}(i)$-linear span is dense, the functions $\lambda x + \lambda' y$ and the predicates being the restrictions of the operations, the inner product and the norm of that space. A model of T_0 is standard if and only if that subset is the whole ball; a nonstandard model of T_0 therefore lacks some of the vectors of the ball and contains none that the ball lacks. In a nonstandard model of T_1 or of T, the sequence sort, which Lemma 6.3 does not concern, may fail to be the closed unit ball of a complex Hilbert space, whether or not the Hilbert sort is the whole ball.

Standardness is a hypothesis of some of our theorems, not a condition imposed by an axiom. The theories T_0, T_1 and T are countable, in countable vocabularies, and they have models, by Theorem A(2) and by taking reducts; the downward Löwenheim–Skolem theorem therefore gives them models whose vector sorts are countable, and such models are not standard, since the closed unit ball of a nonzero Hilbert space is uncountable. The countable models of T nevertheless fall under Theorem C, whereas the conclusions of Theorem A(1) and Theorem B fail in them (see Remark 11.2).

Historical note. Carol Karp, a student of Henkin's, wrote her dissertation of 1959 on languages with expressions of infinite length and published a book on them in 1964 [Kar64]. The model existence theorem for $L_{\omega_1,\omega}$, which builds a model from a consistency property in the manner of Henkin's completeness proof, is due to Makkai [Mak69], and Keisler's book of 1971 proves its theorems through consistency properties [Kei71]. Model theory with truth values in the unit interval goes back to Chang's work on the infinite-valued logic of Łukasiewicz [Cha58, Cha61]; the continuous model theory of Chang and Keisler of 1966 took its truth values in a compact Hausdorff space [CK66]. Henson introduced the logic of positive bounded formulas in 1976 to study a Banach space through its nonstandard hull [Hen76], and Ben Yaacov, Berenstein, Henson and Usvyatsov set out the model theory of metric structures in continuous logic [BYBHU08]. Ben Yaacov and Iovino introduced an infinitary continuous logic for model-theoretic forcing [BYI09], and Eagle proved the omitting types theorem for infinitary $[0,1]$-valued logic [Eag14]. Keisler's real-valued logic [Kei23] gives no metric a priori, and the logic of this paper is the infinitary extension of it due to the first author [Aub].

6. The axioms

In this section we state the axioms of the theory T of Theorem A, and of the theories T_0 and T_1 contained in it. Taken together, the axioms say that the constants e_n form an orthonormal basis; that the elements of the Solèr sort determine orthogonal projections acting on the Hilbert sort, ordered and combined as closed subspaces are; that atoms correspond to unit vectors up to a phase, that is, up to a complex factor of modulus one; that the closed span of the entries of each element of the sequence sort corresponds, through span, to an element of the Solèr sort; and that μ is σ-additive. They come in eight groups: one for each of the three sorts, namely (H), (OL) and (K), the last of which also governs the functions ι_n and κ_n; four for the other symbols that connect the sorts, namely (P) and (Sub) for the projection π and its interplay with the lattice operations, (At) for at and rep, and (Sp) for span; and one, (M), for the measure.

Free variables are universally quantified: the variables x, y, z range over $\mathbf{H}$, the variables s, t over $\mathbf{K}$, and the variables a, b, c over $\mathbf{L}$. The letters λ, λ' (ranging over $\mathbb{Q}(i)$, with $|\lambda|+|\lambda'| \leq 1$) and n, m, k, N (ranging over $\mathbb{N}$) are not variables of the logic: an axiom in which they occur free stands for countably many axioms, one for each value; where they are bound, they index finite sums and countable suprema and infima. Readings in words, in italics, follow the axioms where useful; they describe what the axioms say in standard models.

(H) The Hilbert sort.

(H1) *Inner product space.* For any two terms built from variables by the functions $\lambda x+\lambda' y$ that denote the same $\mathbb{Q}(i)$-linear combination of the variables, the equation between them; the conditions $\langle \lambda x + \lambda' y \mid z\rangle = \lambda\langle x \mid z\rangle + \lambda'\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \overline{\langle x \mid y\rangle}$, each written as two real conditions; and $\|x\| = \sqrt{\max(0, \mathrm{Re}\langle x \mid x\rangle)}$.

(H2) *Separability.* $\langle e_n \mid e_m\rangle = \delta_{nm}$, where δ_{nm} is the Kronecker delta, and

$$\|x\|^2 = \sup_N \min\Big(1, \sum_{n<N} |\langle x \mid e_n\rangle|^2\Big).$$

The constants e_n are orthonormal and satisfy Parseval's identity.

(OL) The Solèr sort.

(OL1) The equations of ortholattices:

$$a \wedge b = b \wedge a, \quad a \wedge (b \wedge c) = (a \wedge b) \wedge c, \quad a \wedge (a \vee b) = a,$$
$$a \vee b = b \vee a, \quad a \vee (b \vee c) = (a \vee b) \vee c, \quad a \vee (a \wedge b) = a,$$
$$a \wedge 0 = 0, \quad a \vee 1 = 1, \quad a^{\perp\perp} = a, \quad (a \wedge b)^{\perp} = a^{\perp} \vee b^{\perp}, \quad a \wedge a^{\perp} = 0.$$

(OL2) *The discrete predicate is equality.*

$$\min\big(d(a,b),\, 1 - d(a,b)\big) = 0, \qquad d(a,a) = 0, \qquad |d(a,c) - d(b,c)| \le d(a,b);$$
$$\max\big(d(a \wedge c, b \wedge c),\, d(a \vee c, b \vee c),\, d(a^{\perp}, b^{\perp})\big) \le d(a,b);$$
$$\max\big(\|\tfrac{1}{2}\pi(a,x) - \tfrac{1}{2}\pi(b,x)\|,\, \|\tfrac{1}{2}\mathrm{rep}(a) - \tfrac{1}{2}\mathrm{rep}(b)\|\big) \le d(a,b);$$
$$|\mu(a) - \mu(b)| \le 3\, d(a,b).$$

The predicate d takes only the values 0 *and* 1 *and is a pseudometric, and the lattice operations, the projections, the function* rep *and the measure respect it.*

(P) Each element of the Solèr sort determines a projection.

(P1) $\pi(a, \lambda x + \lambda' y) \doteq \lambda \pi(a,x) + \lambda' \pi(a,y)$.

(P2) $\pi(a, \pi(a,x)) \doteq \pi(a,x)$.

(P3) $\langle \pi(a,x) \mid y \rangle = \langle x \mid \pi(a,y) \rangle$.

(P4) *Extensionality.* $d(a,b) \le \sup\limits_{n \ge 1} \min\Big(1,\, n \sup\limits_{x} \big\|\tfrac{1}{2}\pi(a,x) - \tfrac{1}{2}\pi(b,x)\big\|\Big)$.

Two elements with the same projection are equal.

(Sub) The order and the operations of the Solèr sort are those of subspaces.

(Sub0) $\pi(0,x) \doteq 0$ and $\pi(1,x) \doteq x$.

Only the zero vector belongs to 0*; every vector belongs to* 1*.*

(Sub1)

$$\|\tfrac{1}{2}\pi(b, \pi(a,x)) - \tfrac{1}{2}\pi(a,x)\| \le d(a \wedge b, a),$$
$$d(a \wedge b, a) \le \sup_{n \ge 1} \min\Big(1,\, n \sup_{x} \big\|\tfrac{1}{2}\pi(b, \pi(a,x)) - \tfrac{1}{2}\pi(a,x)\big\|\Big).$$

The relation $a \le b$ holds if and only if every member of a is a member of b.

(Sub2) $\inf\limits_{N} \sup\limits_{n \ge N} \big\|\tfrac{1}{2}\pi(a \wedge b, x) - \tfrac{1}{2}(P_a P_b)^n x\big\| = 0$, where $(P_a P_b)^n x$ is the term obtained by applying $\pi(b, \cdot)$ and then $\pi(a, \cdot)$, n times.

The members of $a \wedge b$ are the common members of a and b.

(Sub3) $\inf\limits_{N} \sup\limits_{n \ge N} \big\|\tfrac{1}{4}\pi(a \vee b, x) - \tfrac{1}{4}x + \tfrac{1}{4}(P_{a^\perp} P_{b^\perp})^n x\big\| = 0$, the vector inside the norm being denoted by a term built from the functions $\lambda x + \lambda' y$.

The closed subspace spanned by the members of $a \vee b$ is the closure of the sum of the subspaces spanned by the members of a and of b.

(Sub4) $\tfrac{1}{2}\pi(a^{\perp}, x) \doteq \tfrac{1}{2}x - \tfrac{1}{2}\pi(a,x)$.

The members of $a^{\perp}$ are the vectors orthogonal to all members of a.

Axiom (At3) below needs a formula that recognizes atoms. Let $\chi(a)$ be the formula $\sup_b \min\bigl(d(b,0),\, d(b,a),\, 1-d(a\wedge b,b)\bigr)$; by (OL2), it takes the value 1 if some b lies strictly between 0 and a, and 0 otherwise. Thus a nonzero element a is an atom exactly when $\chi(a)=0$.

(At) Atoms and unit vectors.

(At1) $\pi(\mathrm{at}(x),x)\doteq x$, and $\bigl|\langle\pi(\mathrm{at}(x),y)\mid\pi(\mathrm{at}(x),z)\rangle\bigr|^2=\|\pi(\mathrm{at}(x),y)\|^2\,\|\pi(\mathrm{at}(x),z)\|^2$.
Every vector x belongs to $\mathrm{at}(x)$*, and any two members of* $\mathrm{at}(x)$ *are linearly dependent.*

(At2) $\pi(a,\mathrm{rep}(a))\doteq\mathrm{rep}(a)$, and $\bigl|\,\|\mathrm{rep}(a)\|-1\,\bigr|\le 1-d(a,0)$.
The vector $\mathrm{rep}(a)$ *belongs to a, and it is a unit vector when $a\neq 0$.*

(At3) $d(\mathrm{at}(\mathrm{rep}(a)),a)\le\max\bigl(\chi(a),\,1-d(a,0)\bigr)$.
Every atom a equals $\mathrm{at}(\mathrm{rep}(a))$.

(At4) $\|x\|\,\|y\|-|\langle x\mid y\rangle|\le d(\mathrm{at}(x),\mathrm{at}(y))$, and

$$d(\mathrm{at}(x),\mathrm{at}(y))\le\sup_{n\ge1}\min\Bigl(1,\,n\max\bigl(0,\,\|x\|\,\|y\|-|\langle x\mid y\rangle|+\bigl|1-\|x\|\bigr|+\bigl|1-\|y\|\bigr|\bigr)\Bigr).$$

Two unit vectors determine the same atom if and only if one is a multiple of the other by a complex number of modulus one.

(At5) $\mathrm{at}(0)=0$, and $d(\mathrm{at}(x),\mathrm{at}(y))\le\sup_{n\ge1}\min\bigl(1,\,n\|\frac12x-\frac12y\|\bigr)$.

(K) The sequence sort.

(K1) The axioms (H1), written for $\mathbf{K}$.

(K2) $\iota_n(\lambda x+\lambda' y)\doteq\lambda\iota_n x+\lambda'\iota_n y$ and $\kappa_n(\lambda s+\lambda' t)\doteq\lambda\kappa_n s+\lambda'\kappa_n t$; $\langle\iota_n x\mid\iota_m y\rangle=\delta_{nm}\langle x\mid y\rangle$; and $\langle\kappa_n s\mid x\rangle=\langle s\mid\iota_n x\rangle$.

(K3) $\|s\|^2=\sup_N\min\Bigl(1,\,\sum_{n<N}\|\kappa_n s\|^2\Bigr)$.
A sequence is determined by its entries, and its norm is the ℓ^2 norm of its entries.

(K4) $\kappa_n\varepsilon\doteq 2^{-n-1}e_n$.

(Sp) Spans.

(Sp1) $\pi(\mathrm{span}(s),\kappa_n s)\doteq\kappa_n s$.
Every entry of s belongs to $\mathrm{span}(s)$.

(Sp2) $\|\pi(\mathrm{span}(s),x)\|^2\le\sup_{m\ge1}\min\Bigl(1,\,m\sup_N\min\Bigl(1,\,\sum_{n<N}|\langle x\mid\kappa_n s\rangle|^2\Bigr)\Bigr)$.
A vector orthogonal to every entry of s is orthogonal to every member of $\mathrm{span}(s)$.

(Sp3) $d(\mathrm{span}(s),\mathrm{span}(t))\le\sup_{n\ge1}\min\bigl(1,\,n\|\frac12s-\frac12t\|\bigr)$.

The axiom of σ-additivity concerns sequences whose entries are mutually orthogonal, and it needs a formula that measures how far the entries of a sequence are from being mutually orthogonal. Let $\theta(s)$ be the formula $\sup_N\min\bigl(1,\sum_{n,m<N,\,n\neq m}|\langle\kappa_n s\mid\kappa_m s\rangle|\bigr)$; it takes values in $[0,1]$ and vanishes exactly when the entries $\kappa_n s$ of s are mutually orthogonal.

(M) The measure.

(M1) $0 \leq \mu(a) \leq 1$.
(M2) $\mu(1) = 1$.
(M3) *σ-additivity.*

$$\Big|\, \mu(\mathrm{span}(s)) - \sup_N \min\Big(2, \sum_{n<N} \mu(\mathrm{at}(\kappa_n s))\Big)\Big| \leq \sup_{k\geq 1} \min\big(2,\, k\,\theta(s)\big).$$

If the entries of s are mutually orthogonal, the value of μ at $\mathrm{span}(s)$ is the sum of its values at the elements $\mathrm{at}(\kappa_n s)$.

Definition 6.1. The theory T_0 consists of the axioms (H), (OL), (P), (Sub) and (At), without the clause on μ in (OL2), in the vocabulary $\mathcal{L}_0$ obtained from $\mathcal{L}$ by removing the sequence sort with its constant ε, the functions ι_n, κ_n and span, and the predicate μ. The theory T_1 consists of T_0 together with (K) and (Sp), in the vocabulary $\mathcal{L}$ without μ; the theory T consists of T_1 together with (M) and the clause on μ in (OL2).

Lemma 6.2. *In a model of T_0, two elements a, b of the Solèr sort are equal if and only if $d(a,b) = 0$, and two vectors x, y are equal if and only if $\|\frac{1}{2}x - \frac{1}{2}y\| = 0$; the same holds in the sequence sort of a model of T_1.*

Proof. By (H1) and (H2), the form is positive semidefinite on the Hilbert sort. Indeed, if $\mathrm{Re}\langle x \mid x\rangle < 0$ for some x, then $\|x\| = 0$ by (H1), and (H2) gives $\langle x \mid e_n\rangle = 0$ for every n; applied to $\frac{1}{2}x + \frac{1}{2}e_0$, axiom (H2) then gives $\frac{1}{4} = \max\big(0, \frac{1}{4}\,\mathrm{Re}\langle x \mid x\rangle + \frac{1}{4}\big)$, which is absurd. An argument of the same kind, which uses (K1)–(K3), the positivity just proved and the vector $\frac{1}{2}s + \frac{1}{2}\iota_0 e_0$, shows that the form is positive semidefinite on the sequence sort as well. The Cauchy–Schwarz inequality therefore holds for the predicates. With it, axioms (H1), (K1), (OL2), (P1)–(P3), (At5), (K2) and (Sp3) show that every atomic formula takes the same values at a and b when $d(a,b) = 0$, and at x and y when $\|\frac{1}{2}x - \frac{1}{2}y\| = 0$; since models are reduced, the elements in question are equal. ☺

Lemma 6.3. *Let $\mathcal{M}$ be a model of T_0. There is a complex Hilbert space $\bar{E}$ such that the Hilbert sort of $\mathcal{M}$ can be identified with a subset Y of the closed unit ball of $\bar{E}$ whose $\mathbb{Q}(i)$-linear span E is dense, the functions $\lambda x + \lambda' y$ and the predicates being the restrictions of the operations, the inner product and the norm of $\bar{E}$. For every element a of the Solèr sort, the map $\pi(a,\cdot)$ extends uniquely to an orthogonal projection $\bar{P}_a$ of $\bar{E}$. If $\mathcal{M}$ is standard, then Y is the whole closed unit ball of $\bar{E}$.*

Proof. The idea is to recover a vector space from the unit ball by formal multiples. Each pair (k,x), with $k \geq 1$ an integer and $x \in Y$, stands for kx; we put $(k,x) \sim (l,y)$ when $\|\frac{1}{2l}x - \frac{1}{2k}y\| = 0$, and define the sum by $(k,x) + (l,y) = (2kl, \frac{1}{2l}x + \frac{1}{2k}y)$ and the form by $\langle (k,x) \mid (l,y)\rangle = kl\langle x \mid y\rangle$. Multiplication by elements of $\mathbb{Q}(i)$ is defined similarly. Axiom (H1), together with the positivity of the form established in the proof of Lemma 6.2, makes the quotient a $\mathbb{Q}(i)$-vector space E with a positive definite hermitian form, and Lemma 6.2 makes the map from Y to E injective; we let $\bar{E}$ be the completion of E. By (P1)–(P3), the map $\pi(a,\cdot)$ extends to a $\mathbb{Q}(i)$-linear idempotent map Q_a on E, symmetric with respect to the form; hence $\|Q_a x\|^2 = \langle x \mid Q_a x\rangle \leq \|x\|\|Q_a x\|$, the map Q_a is contractive, and it extends by continuity to a self-adjoint idempotent $\bar{P}_a$ of $\bar{E}$. In a standard model, the set Y is the closed unit ball of a Hilbert space, which is complete and spanned by Y; that space is therefore $\bar{E}$. ☺

Historical note. In the continuous logic of Ben Yaacov, Berenstein, Henson and Usvyatsov, the unit ball of a Banach space carries the maps $(x, y) \mapsto \alpha x + \beta y$ with $|\alpha| + |\beta| \leq 1$ as functions, and a Hilbert space carries in addition its inner product as a predicate [BYBHU08]; the Hilbert sort follows this practice. Von Neumann's alternating projection theorem, on which axioms (Sub2) and (Sub3) rest, comes from lecture notes that he distributed in mimeographed form in 1933. The notes were printed in 1950 as the second volume of his *Functional Operators* [vN50].

7. Standard models of T_0: pre-Solèr lattices

In this section we study the standard models of T_0, the theory without the sequence sort and without the measure. We show that the Solèr sort of such a model is a pre-Solèr lattice, that is, that it satisfies every condition in the definition of a Solèr lattice except, possibly, completeness. We also show that the map sending each of its elements to the closed subspace spanned by the members of that element embeds it in the lattice of closed subspaces of the Hilbert space whose closed unit ball is the Hilbert sort, as a sub-ortholattice that is join-dense and meet-dense in the order-theoretic sense of Section 1. Finally, we show that Gleason's theorem already holds at this level: the σ-additive states on the Solèr sort, defined below, are given by density operators.

Throughout this section, we fix a standard model $\mathcal{M}$ of T_0. We write H for the Hilbert space whose closed unit ball is the Hilbert sort of $\mathcal{M}$ (see Definition 5.2), and $\mathbf{L}^{\mathcal{M}}$ for the Solèr sort of $\mathcal{M}$. For each element a of $\mathbf{L}^{\mathcal{M}}$, we write P_a for the projection $\bar{P}_a$ of Lemma 6.3, which extends $\pi(a, \cdot)$, and S_a for its range; this range is the closed subspace of H spanned by the members of a, the vectors x of the unit ball with $\pi(a, x) = x$.

Proposition 7.1. *The space H is a separable infinite-dimensional complex Hilbert space, and $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H.*

Proof. By (H2), the vectors e_n are orthonormal, and Parseval's identity puts every vector of the unit ball, and hence every vector of H, in their closed span. Thus $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H, which is therefore separable and infinite-dimensional; it is complex by Definition 5.2. ☺

We next show that the map $a \mapsto S_a$ respects the order and the operations of the Solèr sort. Axioms (Sub2) and (Sub3) describe meets and joins through von Neumann's alternating projection theorem, which reaches the projection onto the intersection of two closed subspaces by projecting alternately onto each.

Theorem 7.2 (von Neumann's alternating projection theorem [vN49, Lemma 22]**).** *Let U and V be closed subspaces of a Hilbert space, and let P_U, P_V and $P_{U\cap V}$ be the orthogonal projections onto U, V and $U \cap V$. Then, for every vector x of the space, the sequence $\big((P_U P_V)^n x\big)_{n\in\mathbb{N}}$ converges in norm to $P_{U\cap V} x$.*

Proposition 7.3. *The map $\tau\colon a \mapsto S_a$ is an embedding of ortholattices from $\mathbf{L}^{\mathcal{M}}$ into $L(H)$; that is, it is injective and satisfies*

$$S_0 = \{0\}, \quad S_1 = H, \quad S_{a\wedge b} = S_a \cap S_b, \quad S_{a\vee b} = \overline{S_a + S_b}, \quad S_{a^\perp} = S_a^\perp.$$

Moreover, the relation $a \leq b$ holds if and only if $S_a \subseteq S_b$.

Proof. Axiom (Sub0) gives the first two equalities, and axiom (Sub4) gives $P_{a^\perp} = \mathrm{Id} - P_a$, whose range is $S_a^\perp$. For meets and joins we combine the axioms with Theorem 7.2: axiom (Sub2) shows that $P_{a\wedge b} = P_{S_a\cap S_b}$, and axiom (Sub3) that $P_{a\vee b} = \mathrm{Id} - P_{S_a^\perp \cap S_b^\perp}$, whose range is $(S_a^\perp \cap S_b^\perp)^\perp = \overline{S_a + S_b}$. Injectivity follows from extensionality: if $S_a = S_b$, then $P_a = P_b$,

and axiom (P4) gives $a = b$. Finally, the order is read off from axiom (Sub1), which shows that $a \wedge b = a$ exactly when $P_b P_a = P_a$, that is, when $S_a \subseteq S_b$. ☺

Proposition 7.4. *For every nonzero x in the unit ball, the subspace $S_{\mathrm{at}(x)}$ is $\mathbb{C}x$. The atoms of $\mathbf{L}^{\mathcal{M}}$ are exactly the elements $\mathrm{at}(x)$ with $x \neq 0$, and $x \mapsto \mathrm{at}(x)$ induces a bijection from the unit sphere of H, modulo multiplication by complex numbers of modulus one, onto the set of atoms of $\mathbf{L}^{\mathcal{M}}$, whose inverse sends an atom a to the class of $\mathrm{rep}(a)$.*

Proof. By (At1), the vector x lies in $S_{\mathrm{at}(x)}$, and any two vectors of $S_{\mathrm{at}(x)}$ attain equality in the Cauchy–Schwarz inequality, hence are linearly dependent; therefore $S_{\mathrm{at}(x)} = \mathbb{C}x$. Since this ray is an atom of $L(H)$ and τ is an order embedding, the element $\mathrm{at}(x)$ is an atom of $\mathbf{L}^{\mathcal{M}}$. If a is an atom of $\mathbf{L}^{\mathcal{M}}$, then axiom (At2) shows that $\mathrm{rep}(a)$ is a unit vector, and axiom (At3) gives $a = \mathrm{at}(\mathrm{rep}(a))$. For unit vectors x and y, axiom (At4) gives $\mathrm{at}(x) = \mathrm{at}(y)$ if and only if $|\langle x \mid y\rangle| = 1$, that is, if and only if y is a multiple of x by a complex number of modulus one. ☺

Theorem 7.5. *Let $\mathcal{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathcal{M}}$, let S_a be the closed subspace of H spanned by the members of a. Then $\mathbf{L}^{\mathcal{M}}$ is a pre-Solèr lattice, and the image of the map $\tau\colon a \mapsto S_a$ is join-dense and meet-dense in the lattice $L(H)$ of closed subspaces of H, which is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathcal{M}}$. The image of τ contains every finite-dimensional subspace of H and the orthogonal complement of each, and the map τ sends every join that exists in $\mathbf{L}^{\mathcal{M}}$ to the corresponding join in $L(H)$.*

Proof. We begin with the finite-dimensional subspaces and then verify, condition by condition, that $\mathbf{L}^{\mathcal{M}}$ is a pre-Solèr lattice. If F is a finite-dimensional subspace with a basis $x_1, \dots, x_k$ of unit vectors, then $\tau(\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k)) = \overline{F} = F$ by Propositions 7.3 and 7.4, and the orthogonal complement $F^\perp$ is $\tau\big((\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k))^\perp\big)$. Condition (OM) holds because $\tau(\mathbf{L}^{\mathcal{M}})$ is a sub-ortholattice of the orthomodular lattice $L(H)$. For (A), let a be a nonzero element of $\mathbf{L}^{\mathcal{M}}$; the atoms below a are the elements $\mathrm{at}(x)$ with $x \in S_a$ and $0 < \|x\| \leq 1$, and any upper bound b of them in $\mathbf{L}^{\mathcal{M}}$ satisfies $S_b \supseteq S_a$, hence $b \geq a$; thus a is their join. For (CP), let a be an element of $\mathbf{L}^{\mathcal{M}}$ and $\mathrm{at}(x)$ an atom not below a; then $x \notin S_a$, and Proposition 7.3 and Lemma 2.7 give $\tau(a \vee \mathrm{at}(x)) = S_a + \mathbb{C}x$, in which S_a has codimension one; since τ is an order embedding, no element lies strictly between a and $a \vee \mathrm{at}(x)$. Condition (I) holds because the join of two distinct atoms corresponds to a plane, which contains infinitely many rays, all in $\tau(\mathbf{L}^{\mathcal{M}})$. The atoms $\mathrm{at}(e_n)$ give (D), and condition (S) holds with $p_n = \mathrm{at}(e_n)$ and $c_n = \mathrm{at}(\frac{1}{2}(e_n + e_{n+1}))$, by (1) and the computation following it.

Join-density and meet-density of the image of τ follow from two facts about each closed subspace S of H: it is the join in $L(H)$ of its rays, and it equals $\bigcap\{y^\perp : y \in S^\perp,\ 0 < \|y\| \leq 1\}$, where each $y^\perp = S_{\mathrm{at}(y)^\perp}$ lies in the image of τ. By Proposition 1.5, the lattice $L(H)$ is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathcal{M}}$. Finally, let $A \subseteq \mathbf{L}^{\mathcal{M}}$ have a join c in $\mathbf{L}^{\mathcal{M}}$, and let W be the closed span of $\bigcup_{a \in A} S_a$; clearly $W \subseteq S_c$. If y is a vector of $S_c \cap W^\perp$ with $0 < \|y\| \leq 1$, then every $a \in A$ lies below $\mathrm{at}(y)^\perp$, hence so does c, and y is orthogonal to itself, which is absurd. Since $S_c = W + (S_c \cap W^\perp)$, it follows that $S_c = W$. ☺

Example 7.6. Let L_{fc} (for "finite dimension or finite codimension") be the set of finite-dimensional subspaces of ℓ^2 and their orthogonal complements. Take the closed unit ball of ℓ^2 as Hilbert sort, the standard unit vectors as the constants e_n, and L_{fc}, with its lattice operations and orthocomplementation, as Solèr sort. Let d be the discrete metric, with

$d(S, S') = 1$ when $S \neq S'$ and $d(S, S) = 0$; put $\pi(S, x) = P_S x$ and $\mathrm{at}(x) = \mathbb{C}x$; and let $\mathrm{rep}(S)$ be a unit vector chosen in S for each $S \neq \{0\}$, and let $\mathrm{rep}(\{0\})$ be 0. The result is a standard model of T_0 whose Solèr sort L_{fc} is not complete.

Gleason's theorem, in the form we need, concerns two notions: density operators, by which quantum mechanics represents the states of a system, and σ-additive states on an ortholattice. A *density operator* on H is a positive operator ρ whose trace $\operatorname{Tr}\rho = \sum_k \langle \rho u_k \mid u_k \rangle$, computed in any orthonormal basis $(u_k)_{k\in\mathbb{N}}$ of H, equals 1. A *σ-additive state* on an ortholattice Λ is a map $\nu\colon \Lambda \to [0,1]$ with $\nu(1) = 1$ such that $\nu(\bigvee_{n\in J} a_n) = \sum_{n\in J} \nu(a_n)$ for every family $(a_n)_{n\in J}$ of mutually orthogonal elements of Λ, indexed by a set $J \subseteq \mathbb{N}$, whose join exists in Λ.

Theorem 7.7 (Gleason's theorem for T_0). *Let $\mathcal{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathcal{M}}$, let P_a be the orthogonal projection onto the closed subspace spanned by the members of a. The σ-additive states on $\mathbf{L}^{\mathcal{M}}$ are exactly the maps $a \mapsto \operatorname{Tr}(\rho P_a)$, where ρ ranges over the density operators on H, and each such state determines its density operator.*

Proof. If the elements a_n, for n in a finite or countable set, are mutually orthogonal with join c in $\mathbf{L}^{\mathcal{M}}$, then by Theorem 7.5 the subspace S_c is the closure of $\sum_n S_{a_n}$; hence $P_c x = \sum_n P_{a_n} x$ for every $x \in H$, and $\operatorname{Tr}(\rho P_c) = \sum_n \operatorname{Tr}(\rho P_{a_n})$; moreover $\operatorname{Tr}(\rho P_1) = \operatorname{Tr}\rho = 1$. Conversely, let ν be a σ-additive state; additivity on the pair $1, 0$ gives $\nu(0) = 0$. For a unit vector x, put $f(x) = \nu(\mathrm{at}(x))$. For $a \neq 0$ and an orthonormal basis (u_k) of S_a, which is countable, the atoms $\mathrm{at}(u_k)$ are mutually orthogonal and their join in $\mathbf{L}^{\mathcal{M}}$ is a, because every upper bound b of them satisfies $S_b \supseteq S_a$; hence $\nu(a) = \sum_k f(u_k)$. Taking $a = 1$ shows that f is a *frame function* of weight 1: its sum over every orthonormal basis of H is 1. Since H is separable of dimension at least three, Gleason's theorem [Gle57] gives a unique density operator ρ with $f(x) = \langle \rho x \mid x \rangle$ for every unit vector x, and then $\nu(a) = \sum_k \langle \rho u_k \mid u_k \rangle = \operatorname{Tr}(\rho P_a)$. ☻

⌛ **Historical note.** Gleason proved his theorem in 1957 to answer a question of Mackey's: whether the Born rule is the only way to assign probabilities to the closed subspaces of a Hilbert space [Mac57, Gle57]. Frame functions are the heart of Gleason's proof. He shows first, by means of spherical harmonics, that continuous frame functions come from quadratic forms; then, in the hardest step, that in dimension at least three every frame function is continuous; and finally he reduces the general case to these. Gleason noted that the restriction on the dimension is essential: in dimension two the theorem fails. Cooke, Keane and Moran gave in 1985 an elementary proof, longer than Gleason's but with fewer prerequisites [CKM85].

8. Standard models of T_1 and proofs of Theorems A and B

In this section we show that, in a standard model of T_1, every closed subspace of the Hilbert space whose closed unit ball is the Hilbert sort is spanned by the members of an element of the Solèr sort, and we prove Theorems A and B. The first of these results lies beyond the axioms of T_0, which constrain only the elements that the Solèr sort contains: none of them compels it to contain a given closed subspace of infinite dimension and infinite codimension, as the lattice L_{fc} of Example 7.6 shows. The sequence sort supplies what is missing: every closed subspace of a separable Hilbert space is the closed span of a square-summable sequence of vectors, and the function span sends such a sequence to an element of the Solèr sort whose members span that subspace.

Throughout this section, we fix a standard model $\mathcal{M}$ of T_1; we write H and K for the Hilbert spaces whose closed unit balls are its Hilbert sort and its sequence sort, and $\mathbf{L}^{\mathcal{M}}$ for

its Solèr sort. Since T_1 contains T_0, the results of Section 7 apply to $\mathcal{M}$, and P_a, S_a and τ are as there.

Lemma 8.1. *The maps κ_n extend to bounded linear maps from K to H, and the map $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is a linear isometry of K onto the space $\ell^2(\mathbb{N}, H)$ of sequences $(x_n)_{n\in\mathbb{N}}$ in H with $\sum_n \|x_n\|^2 < \infty$.*

Proof. By (K2), each map κ_n is linear on the unit ball of K, and by (K3) it satisfies $\|\kappa_n s\| \leq \|s\|$; it therefore extends to a bounded linear map from K to H, and (K3) shows that the map $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is isometric. By (K2), we also have $\kappa_m \iota_n x = \delta_{mn} x$, and the maps ι_n extend to linear isometries from H into K with mutually orthogonal ranges. Given $(x_n)_{n\in\mathbb{N}} \in \ell^2(\mathbb{N}, H)$, the partial sums $\sum_{n<N} \iota_n x_n$ therefore form a Cauchy sequence in K; since K is complete, they converge to some s, and $\kappa_n s = x_n$ by the continuity of κ_n. ☺

Lemma 8.2. *For every s in the unit ball of K, the subspace $S_{\mathrm{span}(s)}$ is the closed span of the entries $\kappa_n s$.*

Proof. Let W be that closed span. By (Sp1), the subspace $S_{\mathrm{span}(s)}$ contains every entry of s; hence $W \subseteq S_{\mathrm{span}(s)}$. If $x \perp W$, axiom (Sp2) gives $P_{\mathrm{span}(s)} x = 0$; hence $S_{\mathrm{span}(s)} \subseteq W^{\perp\perp} = W$. ☺

Theorem 8.3. *Let $\mathcal{M}$ be a standard model of T_1, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort, let S_a be the closed subspace of H spanned by the members of a. Then H is a separable infinite-dimensional complex Hilbert space, and the map $\tau\colon a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort of $\mathcal{M}$ onto the lattice $L(H)$ of closed subspaces of H; in particular, that Solèr sort is a Solèr lattice. Conversely, for every separable infinite-dimensional complex Hilbert space H', there is a standard model of T_1 whose Hilbert sort is the closed unit ball of H' and whose Solèr sort is $L(H')$.*

Proof. By Proposition 7.1, the space H is separable and infinite-dimensional. By Proposition 7.3, the map τ is an embedding; it remains to be shown that it is onto. Let S be a closed subspace of H, and let $(u_n)_{n\in\mathbb{N}}$ enumerate an orthonormal basis of S, which is countable by Proposition 7.1, followed by zeros if the basis is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_n s = 2^{-n-1} u_n$ for all n, since $\sum_n 4^{-n-1} < 1$, and Lemma 8.2 gives $S_{\mathrm{span}(s)} = S$.

For the converse, we build the model sort by sort. The Hilbert sort is the closed unit ball of H', and the constants e_n are interpreted as an orthonormal basis $(e_n)_{n\in\mathbb{N}}$ of H'. The Solèr sort is $L(H')$, with d the discrete metric, $\pi(S, x) = P_S x$ and $\mathrm{at}(x) = \mathbb{C}x$; for each $S \neq \{0\}$ we let $\mathrm{rep}(S)$ be a unit vector chosen in S, and we let $\mathrm{rep}(\{0\})$ be 0. The sequence sort is the closed unit ball of $\ell^2(\mathbb{N}, H')$, with $\iota_n x$ the sequence with x in position n and 0 elsewhere, $\kappa_n s$ the entry of s in position n, and $\varepsilon = (2^{-n-1} e_n)_{n\in\mathbb{N}}$. Finally, the function span sends each element s of the sequence sort to the closed span of its entries. Axioms (Sub2) and (Sub3) hold by Theorem 7.2, and the others by construction. ☺

Proof of Theorem A. Let $\mathcal{M}$ be a standard model of T; it is in particular a standard model of T_1, and by Theorem 8.3 the space H is a separable infinite-dimensional complex Hilbert space and τ is an isomorphism of $\mathbf{L}^{\mathcal{M}}$ onto $L(H)$.

Step 1: $\mu(0) = 0$. Recall that the formula θ of Section 6 vanishes exactly at the sequences whose entries are mutually orthogonal. It therefore vanishes at the zero element s of the sequence sort, whose entries, all 0, are mutually orthogonal; moreover, Lemma 8.2 gives

$S_{\mathrm{span}(s)} = \{0\}$, whence $\mathrm{span}(s) = 0$ by the injectivity of τ. Axiom (At5) gives $\mathrm{at}(\kappa_n s) = \mathrm{at}(0) = 0$, and axiom (M3) then gives $\mu(0) = \sup_N \min(2, N\mu(0))$; if $\mu(0)$ were positive, the right side would be 2, contradicting (M1).

Step 2: sums along orthonormal bases. Let a be an element of $\mathbf{L}^{\mathcal{M}}$, and let $(u_k)_{k\in\mathbb{N}}$ enumerate an orthonormal basis of S_a, followed by zeros if it is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_k s = 2^{-k-1}u_k$. Its entries are mutually orthogonal, whence $\theta(s) = 0$; by Lemma 8.2 and the injectivity of τ, the element $\mathrm{span}(s)$ is a. For $u_k \neq 0$, Proposition 7.4 gives $S_{\mathrm{at}(2^{-k-1}u_k)} = \mathbb{C}u_k$, hence $\mathrm{at}(\kappa_k s) = \mathrm{at}(u_k)$; for $u_k = 0$, the element $\mathrm{at}(\kappa_k s)$ is 0, whose measure is 0 by Step 1. Axiom (M3) and $\mu(a) \leq 1$ give

$$\mu(a) = \sum_k \mu(\mathrm{at}(u_k)). \tag{$*$}$$

Step 3: application of Gleason's theorem. For a unit vector x, put $f(x) = \mu(\mathrm{at}(x))$. Applying $(*)$ to $a = 1$ and using (M2), we obtain $\sum_k f(u_k) = 1$ for every orthonormal basis (u_k) of H. Arguing as in the last three sentences of the proof of Theorem 7.7, with μ in place of ν and $(*)$ in place of the decomposition used there, we obtain a unique density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in \mathbf{L}^{\mathcal{M}}$.

The converse. Given a separable infinite-dimensional complex Hilbert space H' and a density operator ρ on H', expand the model of Theorem 8.3 for H' by $\mu(a) = \mathrm{Tr}(\rho P_a)$. The clause on μ in (OL2) holds because d is the discrete metric and $|\mu(a) - \mu(b)| \leq 1$. Axioms (M1) and (M2) hold because $\mathrm{Tr}(\rho P) \in [0,1]$ for every orthogonal projection P and $\mathrm{Tr}\,\rho = 1$. For (M3), suppose that the entries $x_n = \kappa_n s$ are mutually orthogonal. Then $\mathrm{span}(s)$ is their closed span, and, with $\hat{x}_n = x_n/\|x_n\|$, we have $\mathrm{Tr}(\rho P_{\mathrm{span}(s)}) = \sum_{x_n\neq 0}\langle \rho\hat{x}_n \mid \hat{x}_n\rangle = \sum_n \mu(\mathrm{at}(x_n)) \leq 1$; the left side of (M3) is therefore 0. If the entries are not mutually orthogonal, the right side of (M3) is 2, while the left side is at most 2. ☺

The proof of Theorem B rests on the fact that, for a density operator ρ, the map $S \mapsto \mathrm{Tr}(\rho P_S)$ on closed subspaces takes every value in $[0,1]$, unlike a measure with only the values 0 and 1.

Lemma 8.4. *Let H be a separable infinite-dimensional complex Hilbert space and ρ a density operator on H. Then $\{\mathrm{Tr}(\rho P_S) : S \in L(H)\} = [0,1]$.*

Proof. The values 0 and 1 are attained at $\{0\}$ and H. Fix α with $0 < \alpha < 1$. We rotate one vector of a subspace spanned by eigenvectors of ρ continuously, from a subspace whose value under $S \mapsto \mathrm{Tr}(\rho P_S)$ is at least α to one whose value is smaller, and apply the intermediate value theorem. Since ρ is a positive compact operator of trace 1, there are orthonormal eigenvectors $v_1, v_2, \ldots$ (finitely or infinitely many) whose eigenvalues $\eta_1 \geq \eta_2 \geq \cdots > 0$ are the nonzero eigenvalues of ρ, repeated according to multiplicity, with $\sum_i \eta_i = 1$. Write $\eta_{\leq k} = \eta_1 + \cdots + \eta_k$, with $\eta_{\leq 0} = 0$, and choose k with $\eta_{\leq k} < \alpha \leq \eta_{\leq k+1}$. On the unit vectors of $W = \{v_1, \ldots, v_{k+1}\}^{\perp}$, the quantity $\langle \rho x \mid x\rangle$ has infimum 0: either W meets the kernel of ρ in a nonzero vector, or the eigenvalues η_i with $i > k+1$ tend to 0. Choose a unit vector $u \in W$ with $\langle \rho u \mid u\rangle < \alpha - \eta_{\leq k}$. For $0 \leq \vartheta \leq 1$, let w_ϑ be the unit vector $\sqrt{1-\vartheta}\, v_{k+1} + \sqrt{\vartheta}\, u$, and let S_ϑ be the subspace spanned by $v_1, \ldots, v_k$ and w_ϑ. Since $\rho v_{k+1} = \eta_{k+1} v_{k+1}$ and u is orthogonal to $v_1, \ldots, v_{k+1}$, the cross terms vanish, and

$$\mathrm{Tr}(\rho P_{S_\vartheta}) = \eta_{\leq k} + (1-\vartheta)\eta_{k+1} + \vartheta\langle \rho u \mid u\rangle,$$

a continuous function of ϑ that equals $\eta_{\leq k+1} \geq \alpha$ at $\vartheta = 0$ and is smaller than α at $\vartheta = 1$. By the intermediate value theorem, it takes the value α. ☺

Proof of Theorem B. Let $\mathcal{M}$ be a standard model of T, and let H be the Hilbert space whose closed unit ball is its Hilbert sort. By Theorem A, there is a density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort, and the map $a \mapsto S_a$ is onto $L(H)$; by Lemma 8.4, the range of μ is $[0,1]$. Since B meets $[0,1]$, some element a of the Solèr sort satisfies $\mu(a) \in B$, and the value of $\mathbf{1}_B(\mu(a))$ at that element is 1; hence the axiom $\sup_a \mathbf{1}_B(\mu(a)) = 0$ of Theorem B fails in $\mathcal{M}$. ☺

⌛ **Historical note.** Sierpiński proved in 1922 that a measure without atoms takes, on the measurable subsets of a set, every value between 0 and the measure of that set [Sie22]. A two-valued measure, which is what an assignment of truth values amounts to, is the opposite extreme: the whole space is an atom. Lemma 8.4 shows that quantum measures fill $[0,1]$ whatever the density operator, even for a pure state, whose classical counterpart would be a point mass.

9. The proof of Theorem C

In this section we prove Theorem C: in no model of T, standard or not, does the measure μ take only the values 0 and 1. Since the model need not be standard, Theorem A is not available, and the proof uses, besides the constants e_n and ε, only the eighteen vectors of Table 1, which are built from four of the e_n.

We argue by contradiction, and fix throughout this section a model $\mathcal{M}$ of T in which μ takes only the values 0 and 1. Let $\bar{E}$ be the complex Hilbert space of Lemma 6.3, in which the Hilbert sort of $\mathcal{M}$ is a subset Y of the closed unit ball whose $\mathbb{Q}(i)$-linear span is dense, and, for each element a of the Solèr sort, let $\bar{P}_a$ be the orthogonal projection of $\bar{E}$ that extends $\pi(a,\cdot)$. By (H2), the vectors e_n form an orthonormal basis of $\bar{E}$: Parseval's identity puts every vector of Y, and hence every vector of $\bar{E}$, in their closed span. The proof rests on three facts that hold in every model of T.

Lemma 9.1. *Let $\mathcal{M}$ be a model of T, standard or not, and let $\bar{E}$ and $\bar{P}_a$ be as in Lemma 6.3. Then the following hold:*

(1) For every nonzero x in the Hilbert sort and every $\lambda \in \mathbb{Q}(i)$ with $0 < |\lambda| \leq 1$, we have $\mathrm{at}(\lambda x) = \mathrm{at}(x)$, and $\bar{P}_{\mathrm{at}(x)}$ is the projection onto $\mathbb{C}x$.

(2) Let s be an element of the sequence sort, and let F be the closed span of its entries in $\bar{E}$. If every e_n lies in F or in $F^\perp$, then $\bar{P}_{\mathrm{span}(s)}$ is the projection onto F.

(3) The measure satisfies $\mu(0) = 0$, and $\mathrm{span}(\varepsilon) = 1$.

Proof. (1) By (At1) and continuity, the range of $\bar{P}_{\mathrm{at}(x)}$ contains x, and any two of its vectors are linearly dependent; hence it is $\mathbb{C}x$, and likewise for λx. The two projections coincide, and axiom (P4) gives $\mathrm{at}(\lambda x) = \mathrm{at}(x)$.

(2) By (Sp1) and linearity, the projection $\bar{P}_{\mathrm{span}(s)}$ fixes every entry of s; hence its range contains F. If $e_n \in F^\perp$, then the vector e_n is orthogonal to every entry, and axiom (Sp2) gives $\bar{P}_{\mathrm{span}(s)} e_n = 0$. The vectors e_n that lie in $F^\perp$ span a dense subspace of $F^\perp$, because the e_n form an orthonormal basis and each lies in F or in $F^\perp$; by continuity, the projection $\bar{P}_{\mathrm{span}(s)}$ vanishes on $F^\perp$, and its range is F.

(3) For $s = 0$, every vector of the Hilbert sort is orthogonal to the entries, and axiom (Sp2) gives $\bar{P}_{\mathrm{span}(0)} = 0$, which is $\bar{P}_0$ by (Sub0); axiom (P4) gives $\mathrm{span}(0) = 0$, and the argument of Step 1 in the proof of Theorem A gives $\mu(0) = 0$. For $s = \varepsilon$, the entries are $2^{-n-1} e_n$ by (K4); the space F of part (2) is then $\bar{E}$, and part (2) gives $\bar{P}_{\mathrm{span}(\varepsilon)} = \mathrm{Id}$, which is $\bar{P}_1$ by (Sub0), whence $\mathrm{span}(\varepsilon) = 1$ by (P4). ☺

basis	vectors (coordinates with respect to an orthonormal basis g_0, g_1, g_2, g_3)			
1	$(0,0,0,1)$	$(0,0,1,0)$	$(1,1,0,0)$	$(1,-1,0,0)$
2	$(0,0,0,1)$	$(0,1,0,0)$	$(1,0,1,0)$	$(1,0,-1,0)$
3	$(1,-1,1,-1)$	$(1,-1,-1,1)$	$(1,1,0,0)$	$(0,0,1,1)$
4	$(1,-1,1,-1)$	$(1,1,1,1)$	$(1,0,-1,0)$	$(0,1,0,-1)$
5	$(0,0,1,0)$	$(0,1,0,0)$	$(1,0,0,1)$	$(1,0,0,-1)$
6	$(1,-1,-1,1)$	$(1,1,1,1)$	$(1,0,0,-1)$	$(0,1,-1,0)$
7	$(1,1,-1,1)$	$(1,1,1,-1)$	$(1,-1,0,0)$	$(0,0,1,1)$
8	$(1,1,-1,1)$	$(-1,1,1,1)$	$(1,0,1,0)$	$(0,1,0,-1)$
9	$(1,1,1,-1)$	$(-1,1,1,1)$	$(1,0,0,1)$	$(0,1,-1,0)$

TABLE 1. The configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]: nine orthogonal bases of a four-dimensional space, formed by eighteen vectors, each of which occurs in exactly two bases. In the proof of Theorem C, the vectors $g_0, \dots, g_3$ are four consecutive vectors $e_{4j}, \dots, e_{4j+3}$, for some $j \in \mathbb{N}$, of the orthonormal basis $(e_n)_{n\in\mathbb{N}}$.

Proof of Theorem C. We first find the one vector of the basis $(e_n)_{n\in\mathbb{N}}$ whose atom has measure 1, and then run a parity argument on the bases of Table 1 inside the block of four basis vectors that contains it. The entries of ε are mutually orthogonal, whence $\theta(\varepsilon) = 0$, and Lemma 9.1(1) gives $\mathrm{at}(2^{-n-1}e_n) = \mathrm{at}(e_n)$; by (M2), (K4), (M3) and Lemma 9.1(3), we obtain

$$1 = \mu(1) = \mu(\mathrm{span}(\varepsilon)) = \sum_n \mu(\mathrm{at}(e_n)).$$

As μ takes only the values 0 and 1, there is exactly one $n_0 \in \mathbb{N}$ with $\mu(\mathrm{at}(e_{n_0})) = 1$. Let $j \in \mathbb{N}$ satisfy $4j \le n_0 < 4j+4$, put $g_i = e_{4j+i}$ for $i = 0,1,2,3$, and let F be the span of $g_0, \dots, g_3$.

The basis $g_0, \dots, g_3$ of F contains exactly one vector, e_{n_0}, whose atom has measure 1; we show that every orthogonal basis of F whose vectors have coordinates in $\{0,1,-1\}$ with respect to $g_0, \dots, g_3$ has the same property. Let $\mathcal{B}$ be such a basis, consisting of vectors $v_0, \dots, v_3$. Each vector $\frac{1}{4}v_i$ has coordinates whose absolute values sum to at most 1, and it is therefore denoted by a term built from the constants e_n by the functions $\lambda x + \lambda' y$. Hence the sequence $\sum_{i<4} \iota_i(\frac{1}{16}v_i)$ is an element $s_{\mathcal{B}}$ of the sequence sort, denoted by the term $\frac{1}{2}\big(\frac{1}{2}\iota_0(\frac{1}{4}v_0) + \frac{1}{2}\iota_1(\frac{1}{4}v_1)\big) + \frac{1}{2}\big(\frac{1}{2}\iota_2(\frac{1}{4}v_2) + \frac{1}{2}\iota_3(\frac{1}{4}v_3)\big)$. The entries of $s_{\mathcal{B}}$ are $\frac{1}{16}v_0, \dots, \frac{1}{16}v_3$ and 0, which are mutually orthogonal; hence $\theta(s_{\mathcal{B}}) = 0$. Each e_n lies in F or in $F^{\perp}$, and Lemma 9.1(2) shows that $\bar{P}_{\mathrm{span}(s_{\mathcal{B}})}$ is the projection onto F. For a vector v with coordinates in $\{0,1,-1\}$, we write $\mathrm{at}(v)$ for $\mathrm{at}(\frac{1}{16}v)$; by Lemma 9.1(1), this element does not change when the sign of v is changed. By (P4), all the elements $\mathrm{span}(s_{\mathcal{B}})$ are equal to one element a_F of the Solèr sort, and axiom (M3), Lemma 9.1(1) and $\mu(0) = 0$ give

$$\mu(a_F) = \sum_{v\in\mathcal{B}} \mu(\mathrm{at}(v)).$$

For the basis $g_0, \dots, g_3$ this sum contains $\mu(\mathrm{at}(e_{n_0})) = 1$ and three zeros; hence $\mu(a_F) = 1$, and every such basis $\mathcal{B}$ contains exactly one vector v with $\mu(\mathrm{at}(v)) = 1$.

Apply this to the nine bases of Table 1, whose vectors span pairwise distinct rays. Count the pairs $(\mathcal{B}, v)$ in which $\mathcal{B}$ is one of the nine bases, $v \in \mathcal{B}$ and $\mu(\mathrm{at}(v)) = 1$. Each basis contributes exactly one pair, and the count is 9. On the other hand, each of the eighteen vectors occurs in exactly two bases, and each vector v with $\mu(\mathrm{at}(v)) = 1$ therefore contributes exactly two pairs; the count is even. This contradiction proves the theorem. ☻

The Bell–Kochen–Specker theorem, Theorem C, has a physical reading. A hidden-variable theory of the kind that Kochen and Specker considered assigns to every experimentally verifiable proposition about a quantum system a definite truth value, 1 or 0. In a model of T the propositions are represented by the elements of the Solèr sort, and Theorem C says that no such assignment can serve as the measure μ, that is, satisfy the axioms (M), σ-additivity included. Combined with axiom (M1), the theorem gives the following.

Corollary 9.2 (No hidden variables). *In every model of T, standard or not, some element a of the Solèr sort satisfies $0 < \mu(a) < 1$.*

Proof. By (M1), the measure takes values in $[0, 1]$, and by Theorem C it does not take only the values 0 and 1. ☻

⧖ **Historical note.** Specker announced in 1960 that the closed subspaces of a three-dimensional Hilbert space cannot be mapped into a Boolean algebra in a way that respects the operations on orthogonal subspaces, and said of the proof only that it rests on an elementary geometric argument [Spe60]. Bell derived from Gleason's theorem that no hidden-variable theory of the kind described above is possible, in a paper written in 1964 and published in 1966 [Bel66], and in 1967 Kochen and Specker gave a finite proof with 117 vectors in $\mathbb{R}^3$ [KS67]. The configurations then shrank. Peres found configurations of 33 vectors in $\mathbb{R}^3$ and of 24 in $\mathbb{R}^4$ in 1991 [Per91], Kernaghan one of 20 vectors in $\mathbb{R}^4$ in 1994 [Ker94], and Cabello, Estebaranz and García-Alcaine the eighteen vectors of Table 1 in 1996 [CEGA96]. Peres conjectured that no smaller configuration exists, and Xu, Chen and Gühne proved in 2020 that no Kochen–Specker set, in any dimension, has fewer than eighteen vectors [XCG20, BCG+22]. The titles of the papers of Kernaghan and of Cabello, Estebaranz and García-Alcaine call the result the Bell–Kochen–Specker theorem, a name that records Bell's share and that Theorem C keeps.

10. The model existence game and finite obstructions

By Theorem C, the measure μ takes a value other than 0 and 1 in every model of the theory T of Theorem A, standard or not. In first-order logic, the compactness theorem would turn this into a finite obstruction: some finite part of the theory would already have no models, as in the Kochen–Specker theorem [KS67], in which finitely many vectors suffice. Real-valued $L_{\omega_1,\omega}$ is not compact, but it satisfies the model existence theorem mentioned in Section 5, which the first author [Aub] proved for this logic. In this section we state the model existence theorem as a game, and we use the game to recover finite obstructions where compactness is not available.

The game follows Väänänen's model existence game for two-valued logic [Vää11]. It is played by Olivia, who claims that a given countable theory has no models, and Josh, who claims that it has one. At each round Olivia makes a challenge, and Josh answers it by adding to a growing description of a structure (see Figure 5); Olivia wins as soon as Josh cannot answer without contradicting himself, and Josh wins if that never happens.

We show that the theory has no models exactly when Olivia has a winning strategy, and that every winning strategy of Olivia determines a well-founded tree whose leaves are finite obstructions, that is, finite sets of statements about the values of sentences that no structure

satisfies. This tree branches infinitely only at the moves of Olivia that concern countable suprema and infima, and once Josh's answers to those moves are fixed, every play ends within a bounded number of rounds. The first three subsections depend only on Section 5. In the last subsection, we return to the theory of Theorem C and describe a winning strategy of Olivia whose finite obstructions are configurations of the Kochen–Specker type.

10.1. **The setting.** Until the last subsection, we work with a countable many-sorted vocabulary $\mathcal{L}$ for the logic of Section 5 and a nonempty countable theory T in $\mathcal{L}$; in the last subsection, the vocabulary and the theory are again those of Section 5.2 and of Theorem A. As usual, every sort of a structure is nonempty.

In the game, Josh commits the values of sentences to intervals, and the rules need, for each sentence, an interval fixed in advance that contains all its possible values. Each predicate symbol P comes with a positive integer r_P such that P takes its values in $[-r_P, r_P]$, and each countable supremum $\sup_n \varphi_n$ or infimum $\inf_n \varphi_n$ comes, as part of its syntax, with a positive rational number r such that every φ_n takes its values in $[-r, r]$ in every structure. By induction on formulas, each formula φ receives a positive rational number r_φ such that φ takes its values in the interval $J_\varphi = [-r_\varphi, r_\varphi]$ in every structure: $r_{P(\bar{t})} = r_P$; for a continuous connective f, the number $r_{f(\varphi_1,\dots,\varphi_\ell)}$ is the least positive integer that is at least the maximum of $|f|$ on the box $J_{\varphi_1} \times \cdots \times J_{\varphi_\ell}$; $r_{\sup_x \varphi} = r_{\inf_x \varphi} = r_\varphi$; and $r_{\sup_n \varphi_n}$ and $r_{\inf_n \varphi_n}$ are the number r that comes with the family $(\varphi_n)_{n\in\mathbb{N}}$. These bounds do not change when terms are substituted for free variables.

The axioms enter the game through one of its rules, which is simplest to state when all axioms have the same form. An axiom is a condition $\varphi = 0$ or $\varphi \le 0$ required to hold for all values of its free variables (see Section 5); replacing $\varphi = 0$ by $|\varphi| \le 0$, and then $\varphi \le 0$ by $\sup_{\bar{x}} \varphi \le 0$, where $\bar{x}$ lists the free variables of φ, we may assume that every axiom of T is a condition $\psi \le 0$ for a sentence ψ.

To name the elements of a structure under construction, we add to $\mathcal{L}$, for each sort $\mathbf{S}$, a set $\mathcal{W}_\mathbf{S} = \{w_{\mathbf{S},i} : i \in \mathbb{N}\}$ of new constants of sort $\mathbf{S}$, the *witnesses*; we write $\mathcal{W}$ for the union of the sets $\mathcal{W}_\mathbf{S}$, and $\mathcal{L}(\mathcal{W})$ for the vocabulary obtained by adding the witnesses to $\mathcal{L}$. We use the letters w and w' for witnesses. Constant symbols of $\mathcal{L}$ are treated as function symbols with no arguments, so that the rules below handle them together with the other function symbols. A sentence of $\mathcal{L}(\mathcal{W})$ is *basic* when it has the form $P(w_1, \dots, w_q)$, where P is a predicate symbol and $w_1, \dots, w_q$ are witnesses.

The players build a partial description of a structure whose elements are named by witnesses. Such a description consists of commitments, which say that the value of a sentence lies in a given interval, and records, which say what a function symbol does to the elements named by given witnesses.

Definition 10.1. A *commitment* is an expression $\varphi \in I$, where φ is a sentence of $\mathcal{L}(\mathcal{W})$ and $I = [\alpha, \beta]$ is an interval with rational endpoints $\alpha \le \beta$; the interval may be a single point. A *record* is an expression $F(\bar{w}) \mapsto w'$, where F is a function symbol and $\bar{w}, w'$ are witnesses of the appropriate sorts. A *position* is a finite set $\mathfrak{p}$ of commitments and records that contains at most one record $F(\bar{w}) \mapsto w'$ for each function symbol F and each tuple $\bar{w}$. A witness *occurs* in $\mathfrak{p}$ when it occurs in a commitment or a record of $\mathfrak{p}$. A position $\mathfrak{p}$ is *infeasible* when there is a basic sentence $P(\bar{w})$ such that the intervals I for which $\mathfrak{p}$ contains the commitment $P(\bar{w}) \in I$, together with $[-r_P, r_P]$, have empty intersection.

An infeasible position is contradictory in the most elementary way: it assigns to a basic sentence $P(\bar{w})$ finitely many rational intervals that have no common point in $[-r_P, r_P]$ (see Figure 7).

The rule for continuous connectives does not ask Josh for the exact values of the arguments of a connective: it asks him to place them in small cells of a grid, fine enough that the connective varies by at most a prescribed amount on each cell. Its statement needs three pieces of notation: dyadic grids, a modulus of uniform continuity for each connective, and widened intervals. For $m \in \mathbb{N}$, the *dyadic grid* $\mathcal{G}_m$ is the set of intervals $[i2^{-m}, (i+1)2^{-m}]$, for $i \in \mathbb{Z}$. Let $f\colon \mathbb{R}^\ell \to \mathbb{R}$ be a continuous connective, let $\varphi_1, \dots, \varphi_\ell$ be sentences, and let $k \in \mathbb{N}$ be given. The function f is uniformly continuous on the compact box $J_{\varphi_1} \times \cdots \times J_{\varphi_\ell}$; we write $m(f, \bar{\varphi}, k)$ for the least $m \in \mathbb{N}$ such that $|f(\xi) - f(\xi')| \le 2^{-k}$ for all points ξ, ξ' of the box with $|\xi_j - \xi'_j| \le 2^{-m}$ for $j = 1, \dots, \ell$. For an interval $I = [\alpha, \beta]$ and $k \in \mathbb{N}$, we write $I^{(k)} = [\alpha - 2^{-k}, \beta + 2^{-k}]$.

Definition 10.2 (The model existence game). The game $\mathrm{G}(T)$ is played by Olivia and Josh. It starts at the empty position; at each round, Olivia makes a *challenge* and Josh *answers* it by enlarging the current position $\mathfrak{p}$. When Josh answers a challenge, the *first fresh witness* of a sort $\mathbf{S}$ is the witness $w_{\mathbf{S},i}$ with the least index i that occurs neither in $\mathfrak{p}$ nor in the challenge. In each rule other than (Ax) and (Tot), the commitment named by Olivia belongs to $\mathfrak{p}$, and its interval is written $I = [\alpha, \beta]$.

(Ax): Olivia names an axiom $\psi \le 0$ of T. Josh adds $\psi \in [-r_\psi, 0]$.

(Tot): Olivia names a function symbol F and witnesses $\bar{w}$ of the appropriate sorts. If $\mathfrak{p}$ contains no record $F(\bar{w}) \mapsto w'$, Josh adds such a record, choosing for w' a witness of the sort of the values of F that occurs in $\mathfrak{p}$ or is the first fresh witness of that sort; otherwise he adds nothing.

(Tm): Olivia names a commitment $P(\bar{t}) \in I$, where P is a predicate symbol and $\bar{t}$ is a tuple of terms, together with a term $F(\bar{w})$ that occurs in $\bar{t}$ and in which $\bar{w}$ is a tuple of witnesses (possibly empty). Josh first answers as in (Tot) for F and $\bar{w}$; then, with w' the witness such that the enlarged position contains the record $F(\bar{w}) \mapsto w'$, he adds $P(\bar{t}') \in I$, where $\bar{t}'$ is obtained from $\bar{t}$ by replacing every occurrence of $F(\bar{w})$ by w'.

(Cn): Olivia names a commitment $f(\varphi_1, \dots, \varphi_\ell) \in I$, where f is a continuous connective, and a number $k \in \mathbb{N}$. With $m = m(f, \bar{\varphi}, k)$, Josh chooses intervals $D_1, \dots, D_\ell$ of $\mathcal{G}_m$ such that each D_j meets J_{φ_j} and f maps $(D_1 \cap J_{\varphi_1}) \times \cdots \times (D_\ell \cap J_{\varphi_\ell})$ into $I^{(k)}$, and adds the commitments $\varphi_j \in D_j \cap J_{\varphi_j}$ (see Figure 6).

($\sup \le$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [-r_\varphi, \beta]$.

($\sup \ge$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness $w \in \mathcal{W}_{\mathbf{S}}$ that occurs in $\mathfrak{p}$ or is the first fresh witness of sort $\mathbf{S}$, and adds $\varphi(w) \in [\alpha - 2^{-k}, r_\varphi]$.

($\inf \ge$): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [\alpha, r_\varphi]$.

($\inf \le$): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness w as in ($\sup \ge$), and adds $\varphi(w) \in [-r_\varphi, \beta + 2^{-k}]$.

($\sup_n \le$): Olivia names a commitment $\sup_n \varphi_n \in I$, where $(\varphi_n)_{n \in \mathbb{N}}$ is a family of formulas with common rational bound $r = r_{\sup_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [-r, \beta]$.

($\sup_n \ge$): Olivia names a commitment $\sup_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [\alpha - 2^{-k}, r]$.

($\inf_n \ge$): Olivia names a commitment $\inf_n \varphi_n \in I$, where $(\varphi_n)_{n \in \mathbb{N}}$ is a family of formulas with common rational bound $r = r_{\inf_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [\alpha, r]$.

($\inf_n \leq$): Olivia names a commitment $\inf_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [-r, \beta + 2^{-k}]$.

An answer is *legal* when it obeys the rule and every interval that it adds is nonempty, that is, has its left endpoint at most its right endpoint. Josh *loses* at a round if he has no legal answer to Olivia's challenge, or if his answer makes the position infeasible. A *play* is a finite or infinite sequence of rounds played according to these rules. Olivia wins a play if Josh loses at some round; Josh wins a play if he never loses, in which case the play is infinite. A *strategy* for a player chooses that player's next move from the moves made so far; it is *winning* when that player wins every play in which the strategy is followed, whatever the opponent does.

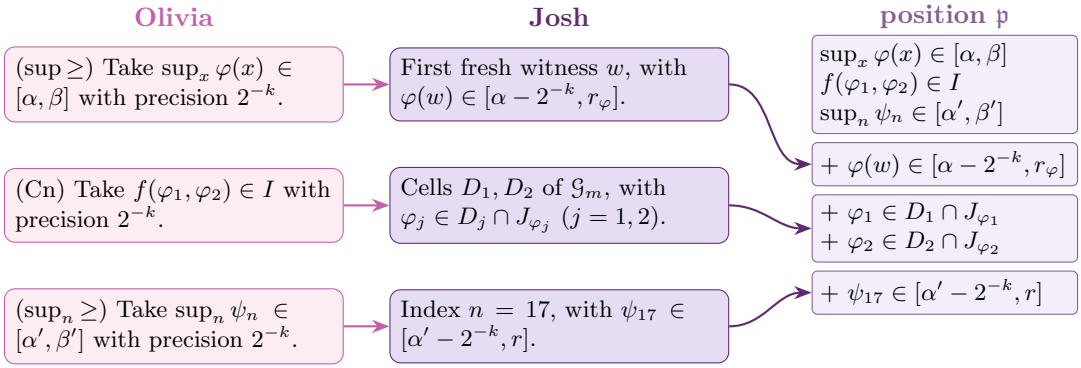


FIGURE 5. Three rounds of the game. Olivia's challenges are on the left and Josh's answers in the middle; the position, on the right, grows by one card at each answer. The first round concerns a quantifier, the second a connective and the third a countable supremum.

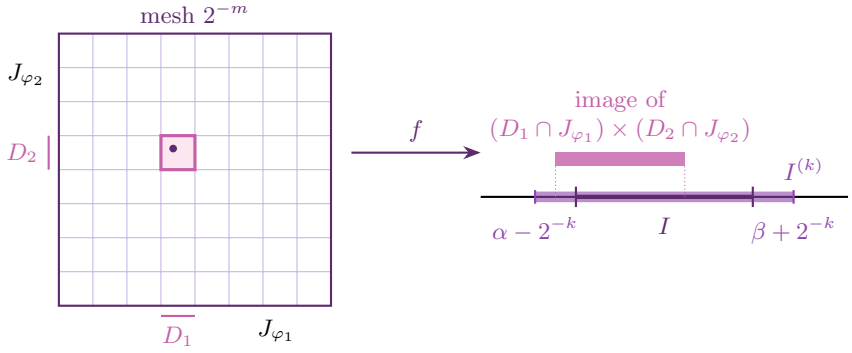


FIGURE 6. The rule (Cn) for a connective f of two variables. Josh must choose a cell $D_1 \times D_2$ of the dyadic grid of mesh 2^{-m} that f maps into the widened interval $I^{(k)}$. A cell that contains the values of φ_1 and φ_2 in a model (the dot) always qualifies, by uniform continuity, although its image may leave I itself.

Josh's answers are of three kinds, according to the number of choices that the rules leave him. Under the rules (Ax), (sup $\leq$), (inf $\geq$), (sup$_n \leq$) and (inf$_n \geq$), Josh has no choice: the answer is determined by the challenge. Under the rules (Tot), (Tm), (Cn), (sup $\geq$) and

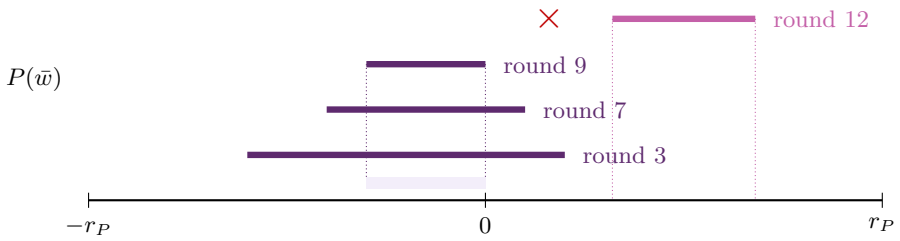


FIGURE 7. An infeasible position: at different rounds, Josh has committed the basic sentence $P(\bar{w})$ to intervals that have no common point.

$(\inf \leq)$, Josh has finitely many choices, since only finitely many witnesses occur in a position and only finitely many intervals of $\mathcal{G}_m$ meet a bounded interval. Only under the rules $(\sup_n \geq)$ and $(\inf_n \leq)$, which concern countable suprema and infima, does Josh choose among infinitely many answers. We call the challenges made under these two rules *countable challenges*, and Josh's answers to them *countable answers*.

10.2. **The model existence theorem as a game.** To extract a model from a play that Josh survives, we need Olivia to have made, sooner or later, every challenge open to her. A challenge is *available* at a round when the rules allow Olivia to make it there, that is, when the commitment that it names, if any, belongs to the position at the start of that round; since positions only grow, a challenge that is available at a round remains available at every later round. Every sentence that occurs in a play is an atomic sentence of $\mathcal{L}(\mathcal{W})$ or is obtained from a subformula of an axiom of T by substituting closed terms of $\mathcal{L}(\mathcal{W})$ for its free variables; hence only countably many challenges are available at each round, and there is at least one, since T is nonempty. A play is *fair* if every challenge that is available at some round is made by Olivia at that round or at a later one. Olivia has a *fair strategy*, one under which every play is fair, whatever Josh does; we call the following one her *bookkeeping*. She fixes a bijection $i \mapsto (j(i), l(i))$ from $\mathbb{N}$ onto $\mathbb{N} \times \mathbb{N}$ with $j(i) \leq i$ for every $i \in \mathbb{N}$, and at each round she enumerates, with repetitions if necessary, the challenges available at that round. At round i she makes the $l(i)$-th challenge of her enumeration for round $j(i)$, which is still available because $j(i) \leq i$.

Theorem 10.3 (The model existence theorem as a game). *Let T be a nonempty countable theory of real-valued $L_{\omega_1,\omega}$ in a countable vocabulary, whose axioms are conditions $\psi \leq 0$ for sentences ψ, and let $\mathrm{G}(T)$ be the game of Definition 10.2. The following conditions are equivalent:*

(1) the theory T has a model;
(2) Josh has a winning strategy in $\mathrm{G}(T)$;
(3) Olivia has no winning strategy in $\mathrm{G}(T)$.

Proof. (1) implies (2). Let $\mathcal{N}$ be a model of T. Josh answers every challenge truthfully in $\mathcal{N}$. To do so, he keeps, along the play, a map h from the witnesses occurring in the current position to $\mathcal{N}$ such that every commitment of the position holds in $\mathcal{N}$ when each witness w is interpreted as $h(w)$, and $F^{\mathcal{N}}(h(\bar{w})) = h(w')$ for every record $F(\bar{w}) \mapsto w'$ of the position. When a challenge names a witness that does not yet occur, Josh first extends h to it arbitrarily; this is possible because the sorts of $\mathcal{N}$ are nonempty. He answers as follows. For (Ax), the added commitment holds because $\mathcal{N}$ is a model of T. For (Tot) and (Tm), if

the position contains no record $F(\bar{w}) \mapsto w'$, Josh adds one with w' the first fresh witness, and puts $h(w') = F^{\mathcal{N}}(h(\bar{w}))$; in either case, the value of $P(\bar{t}')$ in $\mathcal{N}$ equals that of $P(\bar{t})$. For (Cn), let ξ_j be the value of φ_j in $\mathcal{N}$; Josh chooses $D_j \in \mathcal{G}_m$ containing ξ_j. Every point of $(D_1 \cap J_{\varphi_1}) \times \cdots \times (D_\ell \cap J_{\varphi_\ell})$ has coordinates within 2^{-m} of those of $(\xi_1, \ldots, \xi_\ell)$, so f maps it to within 2^{-k} of $f(\xi_1, \ldots, \xi_\ell)$, which lies in I. For $(\sup \leq)$, the value of $\varphi(w)$ is at most the supremum, hence at most β. For $(\sup \geq)$, the supremum is at least α; Josh takes the first fresh witness w and lets $h(w)$ be an element of $\mathcal{N}$ at which φ takes a value at least $\alpha - 2^{-k}$. The remaining rules are handled in the same way; for instance, for $(\sup_n \geq)$, some n gives φ_n a value at least $\alpha - 2^{-k}$. Every answer is legal, since the true values lie in the added intervals, and no position is infeasible, since the value in $\mathcal{N}$ of a basic sentence $P(\bar{w})$ lies in every interval committed for it and in $[-r_P, r_P]$. Thus Josh never loses.

(2) implies (1). Let Josh play a winning strategy against a fair strategy of Olivia. The resulting play is infinite and fair; let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots$ be its positions, indexed by $i \in \mathbb{N}$, and $\mathfrak{p}_\infty$ their union. We define a structure $\mathcal{M}$ for $\mathcal{L}(\mathcal{W})$ whose elements are the witnesses: the universe of sort $\mathbf{S}$ is $\mathcal{W}_{\mathbf{S}}$, and each witness is interpreted as itself. For a function symbol F and witnesses $\bar{w}$, Olivia makes the challenge (Tot) for F and $\bar{w}$ at some round, because the play is fair; hence $\mathfrak{p}_\infty$ contains a record $F(\bar{w}) \mapsto w'$, which is unique because every $\mathfrak{p}_i$ is a position, and we put $F^{\mathcal{M}}(\bar{w}) = w'$. For a basic sentence $P(\bar{w})$, the intervals I for which $\mathfrak{p}_\infty$ contains $P(\bar{w}) \in I$, together with $[-r_P, r_P]$, are compact, and every finite subfamily has a common point because no position $\mathfrak{p}_i$ is infeasible; by the finite intersection property, the whole family has a common point, and we let $P^{\mathcal{M}}(\bar{w})$ be one. No continuity is required of $F^{\mathcal{M}}$ or $P^{\mathcal{M}}$ (see Section 5); hence $\mathcal{M}$ is a structure for $\mathcal{L}(\mathcal{W})$.

We claim that every commitment $\varphi \in I$ of $\mathfrak{p}_\infty$ holds in $\mathcal{M}$, and prove it by induction on the construction of φ and, for atomic sentences, on the number of occurrences of function symbols. For a basic sentence, the claim holds by the choice of $P^{\mathcal{M}}$. A non-basic atomic sentence $P(\bar{t})$ contains a term $F(\bar{w})$ whose arguments $\bar{w}$ are witnesses (possibly none, when F is a constant symbol); by fairness, Olivia made the challenge (Tm) for the commitment and this term, and Josh added $P(\bar{t}') \in I$, which holds by the induction hypothesis; since $F^{\mathcal{M}}(\bar{w})$ is the witness w' that replaces $F(\bar{w})$, the sentences $P(\bar{t})$ and $P(\bar{t}')$ have the same value in $\mathcal{M}$. For $f(\varphi_1, \ldots, \varphi_\ell) \in I$, fairness gives, for every $k \in \mathbb{N}$, intervals $D_1, \ldots, D_\ell$ such that $\mathfrak{p}_\infty$ contains the commitments $\varphi_j \in D_j \cap J_{\varphi_j}$ and f maps $(D_1 \cap J_{\varphi_1}) \times \cdots \times (D_\ell \cap J_{\varphi_\ell})$ into $I^{(k)}$; by the induction hypothesis, the value of $f(\varphi_1, \ldots, \varphi_\ell)$ in $\mathcal{M}$ lies in $I^{(k)}$ for every k, hence in I. For $\sup_x \varphi(x) \in [\alpha, \beta]$, fairness puts in $\mathfrak{p}_\infty$ a commitment $\varphi(w) \in [-r_\varphi, \beta]$ for every witness w of the sort of x, that is, for every element of $\mathcal{M}$ of that sort, and, for every $k \in \mathbb{N}$, a commitment $\varphi(w) \in [\alpha - 2^{-k}, r_\varphi]$ for some witness w; by the induction hypothesis, the supremum lies in $[\alpha, \beta]$. The remaining cases are similar. Every axiom $\psi \leq 0$ of T was named in a challenge (Ax), so $\mathfrak{p}_\infty$ contains $\psi \in [-r_\psi, 0]$, and $\mathcal{M}$ satisfies $\psi \leq 0$.

It remains to pass to a reduced structure. For witnesses w_1, w_2 of the same sort, let $w_1 \sim w_2$ when $\varphi(w_1, \bar{w})$ and $\varphi(w_2, \bar{w})$ have the same value in $\mathcal{M}$ for every atomic formula $\varphi(x, \bar{y})$ of $\mathcal{L}$ and every tuple $\bar{w}$ of witnesses. If $w_1 \sim w_2$, then $F^{\mathcal{M}}(w_1, \bar{w}') \sim F^{\mathcal{M}}(w_2, \bar{w}')$ for every function symbol F and every tuple $\bar{w}'$ of witnesses, since $\varphi(F(x, \bar{y}'), \bar{y})$ is again atomic; the same holds at every argument place of F, and by transitivity, the relation $\sim$ is a congruence for every function symbol, whether or not it is continuous. The quotient $\mathcal{M}/\!\sim$ is reduced, and the quotient map is surjective and preserves the value of every atomic formula; by induction on formulas, it preserves the value of every formula. Hence the reduct of $\mathcal{M}/\!\sim$ to $\mathcal{L}$ is a model of T.

(2) and (3) are equivalent. Regard $\mathrm{G}(T)$ as an infinite game in which, after Josh has lost, the play continues with arbitrary moves. Olivia wins exactly when Josh loses at some finite round; in the language of descriptive set theory, the set of her wins is *open* in the space of infinite sequences of moves, whose basic open sets consist of the sequences that extend a given finite one. The Gale–Stewart theorem [GS53] says that an infinite game of perfect information in which the set of wins of one player is open is *determined*: one of the two players has a winning strategy. Since the two players cannot both have winning strategies, (2) and (3) are equivalent.

Remark 10.4 (Baire category and the Banach–Mazur game). Olivia's bookkeeping can be read as a Baire category argument. Let σ' be a winning strategy for Josh, and let $X_{\sigma'}$ be the set of infinite plays in which Josh follows σ', with the topology whose basic open sets consist of the plays that extend a given finite play. Since the moves range over countable sets, the space $X_{\sigma'}$ is homeomorphic to a closed subspace of the Baire space $\mathbb{N}^{\mathbb{N}}$, the space of sequences of natural numbers with the product of the discrete topologies; it is therefore a *Polish* space, that is, separable and completely metrizable, and the Baire category theorem holds in it. For each challenge and each round, the set of plays in which the challenge, if available at that round, is made at that round or later is open and dense in $X_{\sigma'}$. The fair plays therefore form a dense G_δ subset of $X_{\sigma'}$, that is, a countable intersection of dense open sets, and by the Baire category theorem this subset is nonempty. The same fact can be phrased as a game. In the *Banach–Mazur game* on a space, two players alternately choose nonempty open sets, each contained in the one before, and the second player wins if the intersection of the chosen sets is contained in a given set (see Figure 8). In the Banach–Mazur game on $X_{\sigma'}$ in which the players choose basic open sets and the second player wins if the resulting play is fair, Olivia's bookkeeping is a winning strategy for the second player (compare Oxtoby [Oxt57]). The argument is the analogue of the Baire category argument by which Eagle [Eag14] proved the omitting types theorem for $[0,1]$-valued $L_{\omega_1,\omega}$; in the proof of Theorem 10.3 it reduces to Olivia's bookkeeping, because only countably many challenges are available at each round.

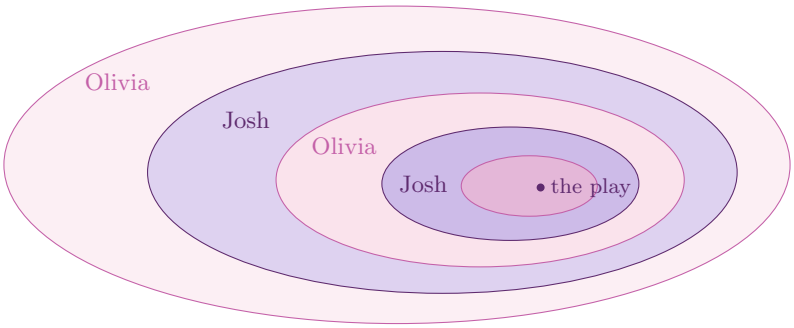


FIGURE 8. The Banach–Mazur view of a play. Olivia (pink) and Josh (purple) move alternately, and each move shrinks the set of plays that extend the current finite play; the intersection of the nested sets is a single play.

10.3. **Finite obstructions.** A winning strategy for Olivia is a certificate that a theory has no models, and its shape shows where finite obstructions suffice. Let σ be a strategy for Olivia. The *tree of* σ, denoted $\mathcal{P}_\sigma$, is the set of finite plays in which Olivia follows σ and Josh has not lost before the last round, ordered by extension. A finite play of $\mathcal{P}_\sigma$ in which Josh has

not lost has as children its extensions by Olivia's next challenge and a legal answer of Josh, and, when Josh has no legal answer to that challenge, its extension by the challenge alone. A *leaf* of $\mathcal{P}_\sigma$ is a finite play with no children, that is, one at which Josh has lost. The strategy σ is winning exactly when $\mathcal{P}_\sigma$ has no infinite branch, that is, when $\mathcal{P}_\sigma$ is *well-founded*. The *rank* of a well-founded tree is defined by recursion: a leaf has rank 0, a node that is not a leaf has as rank the least ordinal greater than the ranks of all its children, and the rank of the tree is the rank of its root. The tree $\mathcal{P}_\sigma$ is countably branching, hence countable; when it is well-founded, its rank is therefore a countable ordinal, and the rank is finite exactly when the plays in which Olivia follows σ have bounded length.

In Corollary 10.5 below, we show that $\mathcal{P}_\sigma$ branches infinitely only at nodes where σ makes a countable challenge; the following notions keep track of Josh's answers to such challenges. The *countable record* of a node of $\mathcal{P}_\sigma$ is the finite sequence of the numbers n that Josh names in his countable answers along the branch from the root to the node, in the order in which he names them. The *skeleton* of σ is the set of countable records of nodes of $\mathcal{P}_\sigma$, ordered by extension; it is a tree of finite sequences of natural numbers.

Corollary 10.5 (Finite obstructions). *Let T be a nonempty countable theory of real-valued $L_{\omega_1,\omega}$ in a countable vocabulary, whose axioms are conditions $\psi \leq 0$ for sentences ψ, and suppose that T has no models. Then Olivia has a winning strategy in $\mathrm{G}(T)$, and every winning strategy σ of Olivia has the following properties:*

(1) The tree $\mathcal{P}_\sigma$ is well-founded: every play in which Olivia follows σ ends after finitely many rounds, at a leaf.

(2) At every leaf, the last position $\mathfrak{p}$ is a finite set of commitments and records, involving finitely many axioms of T and finitely many witnesses, that is satisfied in no structure under any interpretation of the witnesses. Either $\mathfrak{p}$ is infeasible, in which case a basic sentence receives finitely many rational intervals with no common point, or Josh has no legal answer to Olivia's last challenge, which names a commitment of $\mathfrak{p}$ that holds in no structure.

(3) A node of $\mathcal{P}_\sigma$ has infinitely many children only if σ makes a countable challenge at it; every other node has finitely many children.

(4) If every node of $\mathcal{P}_\sigma$ has finitely many children, as is the case when σ makes no countable challenge, and in particular when no axiom of T contains a countable supremum or infimum, then $\mathcal{P}_\sigma$ is finite. In that case there is a number N_σ such that every play in which Olivia follows σ ends within N_σ rounds, and T has a finite subset that has no models.

(5) The skeleton of σ is well-founded, and for every element s of the skeleton there is a number $N_\sigma(s)$ such that every play in which Olivia follows σ and whose countable record is s ends within $N_\sigma(s)$ rounds.

Proof. By Theorem 10.3, Olivia has a winning strategy; let σ be any winning strategy of Olivia. Part (1) holds because σ is winning. For part (2), a position contains finitely many commitments and records, and the argument for "(1) implies (2)" in the proof of Theorem 10.3 shows that a position satisfied in some structure, under some interpretation of the witnesses, is not infeasible and allows Josh a legal answer to every challenge. Part (3) follows from the count of Josh's possible answers in the paragraph after Definition 10.2.

Parts (4) and (5) rest on Kőnig's lemma: an infinite tree in which every node has finitely many children has an infinite branch. For part (4), the tree $\mathcal{P}_\sigma$ is well-founded and, by hypothesis, finitely branching; hence it is finite, and the length of its longest branch gives N_σ. The hypothesis holds, by part (3), when σ makes no countable challenge, and in particular

when no axiom of T contains a countable supremum or infimum, since every sentence that occurs in a play is atomic or is an instance of a subformula of an axiom. It remains to be shown that T has a finite subset that has no models. Let T' be the set of axioms named in challenges (Ax) at nodes of $\mathcal{P}_\sigma$, together with one axiom of T if there are none. In $\mathrm{G}(T')$, every challenge that σ makes at a node of $\mathcal{P}_\sigma$ is allowed by the rules, and Josh has the same answers as in $\mathrm{G}(T)$; hence σ wins $\mathrm{G}(T')$, and the finite theory T' has no models by Theorem 10.3.

For part (5), let s be a finite sequence of natural numbers, and let $\mathcal{P}^s_\sigma$ be the set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of s; it is a subtree of $\mathcal{P}_\sigma$, and every play in which Olivia follows σ and whose countable record is s runs inside it. In $\mathcal{P}^s_\sigma$, a node at which σ makes a countable challenge has at most one child. If Josh has no legal answer to the challenge, its only child is the extension by the challenge alone. Otherwise, its only child is the extension by the countable answer whose number is the next entry of s, unless the countable record of the node is s itself, in which case it has no child in $\mathcal{P}^s_\sigma$. By part (3), every other node of $\mathcal{P}^s_\sigma$ has finitely many children. Since $\mathcal{P}^s_\sigma$ is well-founded, it is finite by Kőnig's lemma, and the length of its longest branch gives $N_\sigma(s)$. Finally, the skeleton is well-founded. If it had an infinite branch $s_1 \subsetneq s_2 \subsetneq \cdots$, the set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of some s_i would form a finitely branching subtree of $\mathcal{P}_\sigma$ containing nodes of every countable record s_i, hence infinite; by Kőnig's lemma it would have an infinite branch, contradicting part (1). ☺

For countable theories of the finitary part of the logic, part (4) of Corollary 10.5 yields the compactness theorem: a countable theory without countable suprema and infima that has no models has a finite subset that has no models. Part (3) locates the possible failures of compactness at the countable challenges: only there, where Josh chooses an index n among infinitely many, can the tree branch infinitely. By part (5), nothing else stands in the way of a uniform bound: the skeleton, a well-founded countably branching tree, records the infinitary shape of the obstruction, and once Josh's countable answers are fixed, the plays have bounded length (see Figures 9 and 10). For a theory with countable suprema and infima, the single finite obstruction of the compact case is thus replaced by a well-founded tree of finite obstructions, one at each leaf.

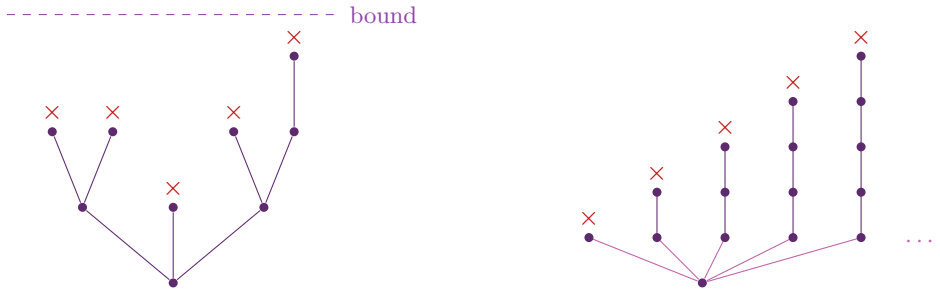


FIGURE 9. Trees grow upward from their roots. Left: a finitely branching tree without infinite branches is finite, by Kőnig's lemma, and its branches have bounded length. Right: a countably branching tree without infinite branches may have branches of every finite length, as in this fan with one branch of each length.

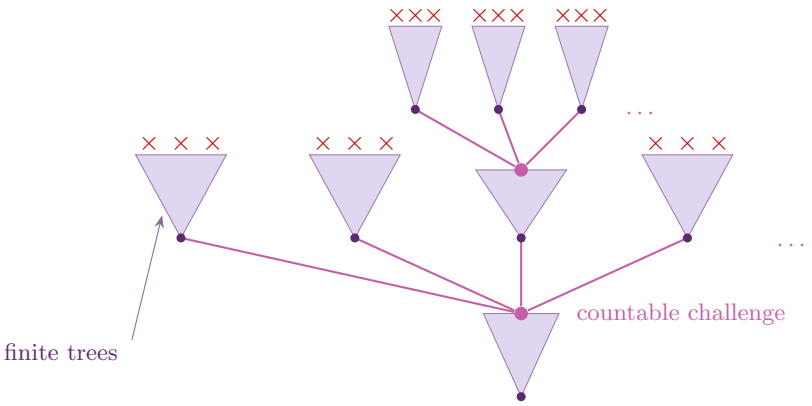


FIGURE 10. The tree of a winning strategy for Olivia, growing upward from its root. At each countable challenge (pink dot) to which Josh has a legal answer, the tree branches infinitely (pink fan), one branch for each number that he may name; the sequences of numbers that he names along the branches form the skeleton. The parts of the tree between countable challenges are finite (shaded), and their leaves, marked $\times$, are finite obstructions.

10.4. **The Bell–Kochen–Specker obstruction.** We now return to the theory T of Theorem A, with its measure μ, and to the axiom of Theorem C, which says that μ takes only the values 0 and 1. In the normal form described in the first subsection, that axiom becomes $\sup_a \bigl|\min\bigl(\mu(a), 1-\mu(a)\bigr)\bigr| \leq 0$; since axiom (M1) makes $\min\bigl(\mu(a), 1-\mu(a)\bigr)$ nonnegative, the absolute value may be dropped. By Theorem C, the theory obtained by adding the resulting axiom to T has no models, and Corollary 10.5 applies to it. We show that the tree of every winning strategy of Olivia for this theory branches infinitely, and we describe a winning strategy whose finite obstructions are configurations of the Kochen–Specker type.

Proposition 10.6. *Let T^{01} be the theory T of Theorem A together with the axiom*

$$\sup_a \min\bigl(\mu(a), 1-\mu(a)\bigr) \leq 0.$$

(1) Olivia has a winning strategy in $\mathrm{G}(T^{01})$; every play in which she follows it ends after finitely many rounds, at a position that is satisfied in no structure.
(2) Every finite subset of T^{01} has a model.
(3) The skeleton of every winning strategy of Olivia in $\mathrm{G}(T^{01})$ has more than one element: in some play in which Olivia follows such a strategy, Josh answers a countable challenge.

Proof. Part (1) follows from parts (1) and (2) of Corollary 10.5, since T^{01} has no models by Theorem C.

For part (2), let T' be a finite subset of T^{01}, and let $S \subseteq \mathbb{N}$ be the finite set of numbers n for which the instance $\kappa_n \varepsilon \doteq 2^{-n-1} e_n$ of axiom (K4) belongs to T'. Consider the following structure. Its Hilbert sort is the closed unit ball of ℓ^2; its Solèr sort is the lattice L_{fc} of Example 7.6, with the interpretations of d, π, at, rep and the constants e_n given there; its sequence sort consists of the finitely supported sequences in the closed unit ball of $\ell^2(\mathbb{N}, \ell^2)$, with ι_n and κ_n as in the proof of Theorem 8.3; and $\mathrm{span}(s)$ is the span of the entries of s, which is finite-dimensional. The measure μ vanishes on the finite-dimensional subspaces and equals 1 on their orthogonal complements, and ε is the finitely supported sequence whose

entry of index n is $2^{-n-1}e_n$ for $n \in S$ and 0 otherwise; its norm is less than 1. The axioms of T_0 hold as in Example 7.6, and the axioms (K1)–(K3) and (Sp) as in the proof of Theorem 8.3, since the span of finitely many vectors is closed; of the instances of (K4), the only axiom in which ε occurs, exactly those with $n \in S$ hold. Axioms (M1) and (M2), the clause on μ in (OL2) and the axiom $\sup_a \min\bigl(\mu(a), 1-\mu(a)\bigr) \leq 0$ hold because μ takes only the values 0 and 1, with $\mu(1) = 1$. For (M3), let s be an element of the sequence sort. If its entries are mutually orthogonal, then $\mathrm{span}(s)$ and the elements $\mathrm{at}(\kappa_n s)$ are finite-dimensional, and both sides of (M3) vanish; otherwise the right side is 2, and the left side is at most 2. The structure is therefore a model of T'.

For part (3), suppose that the skeleton of a winning strategy σ of Olivia had only one element, the empty sequence. Then Josh would give no countable answer at any node of $\mathcal{P}_\sigma$: a node at which σ makes a countable challenge would have only one child, the extension by the challenge alone, and every other node would have finitely many children by Corollary 10.5(3). By Corollary 10.5(4), the theory T^{01} would then have a finite subset without models, contradicting part (2). ☻

Proof of Theorem D. In the model existence game, Olivia is the player who claims that the theory has no models, and the tree of plays of a strategy σ of hers is the tree $\mathcal{P}_\sigma$. Let T' be a countable theory that has no models, with its axioms put in the normal form of the first subsection. Then T' is nonempty, since the empty theory has models, and the assertions of Theorem D about T' are Theorem 10.3 and Corollary 10.5, with T' in place of T. For the theory T' of Theorem C, every finite subset has a model, namely the structure constructed in the proof of Proposition 10.6(2), whose measure takes only the values 0 and 1; the argument for Proposition 10.6(3) then shows that, in the tree of every winning strategy of Olivia, Josh gives a countable answer at some node. Since the interval that Josh adds under the rules $(\sup_n \geq)$ and $(\inf_n \leq)$ does not depend on the number he names, every number is a legal answer at that node, and the tree branches infinitely there. ☻

We describe informally a winning strategy for Olivia that follows the proof of Theorem C (see Figures 11 and 12). It has two phases: the first finds a basis vector whose atom has measure close to 1, and the second runs the parity argument of Theorem C in the block of four basis vectors that contains it. Recall that the constant ε names the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$ built from the constants e_n, which form an orthonormal basis. Olivia begins with the instance of axiom (M3) at ε; from it and from the instances of (H1), (H2), (OL2), (P4), (Sub0), (At1), (At5), (K2), (K4), (Sp1), (Sp2), (M1) and (M2) used in the proofs of Lemma 9.1 and Theorem C, she obtains commitments saying that the sum of the values $\mu(\mathrm{at}(\kappa_n\varepsilon))$ is close to 1. A countable challenge $(\sup_n \geq)$ to the supremum $\sup_N \min\bigl(2, \sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))\bigr)$ in (M3) forces Josh to name a number N for which the finite sum $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ is close to 1. Challenges to the axiom of T^{01} at witnesses that name the elements $\mathrm{at}(\kappa_n\varepsilon)$, for $n < N$, followed by challenges (Cn) and by challenges to (M1), make him commit each value $\mu(\mathrm{at}(\kappa_n\varepsilon))$ to an interval close to 0 or close to 1. Since Olivia chooses the precision of these challenges after Josh has named N (finer than $1/(2N)$, say), the finite sum forces at least one $n_0 < N$ with value close to 1. Through (K4) and the axioms behind Lemma 9.1(1), the element $\mathrm{at}(\kappa_{n_0}\varepsilon)$ is identified with $\mathrm{at}(e_{n_0})$. Further countable challenges arise, here and in the second phase, from the countable suprema in (H2), in the formula θ that precedes (M3) and in the uses of the device (2) in (P4) and (Sp2).

In the second phase, Olivia turns to the block of four basis vectors $e_{4j}, \ldots, e_{4j+3}$ with $4j \leq n_0 < 4j+4$, and challenges the instances of the axioms that the proof of Theorem C uses for the nine bases of Table 1 and for the block itself. Josh's answers commit each of

the eighteen values $\mu(\mathrm{at}(v))$ to an interval close to 0 or close to 1, and each of the nine sums over a basis to an interval close to 1. Whatever finitely many choices Josh makes, the parity argument in the proof of Theorem C shows that these commitments are incompatible. Through the instances of (OL2), (K2) and (P4) that identify the elements $\mathrm{at}(v)$ reached through different bases, Olivia can then drive this incompatibility down to a basic sentence that receives intervals with no common point. Thus the final position of each leaf contains a finite configuration of the Kochen–Specker type inside one block of four basis vectors, the block that Josh's answer N bounds and his finitely many choices of intervals select.

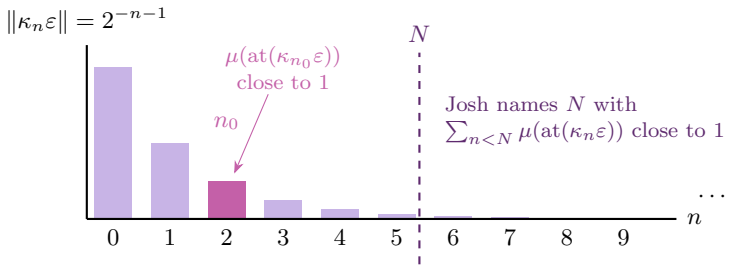


FIGURE 11. The first countable answer. The entries of ε have norms 2^{-n-1}; Josh names N with $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ close to 1, and the axiom of T^{01} then forces some $n_0 < N$ with value close to 1.

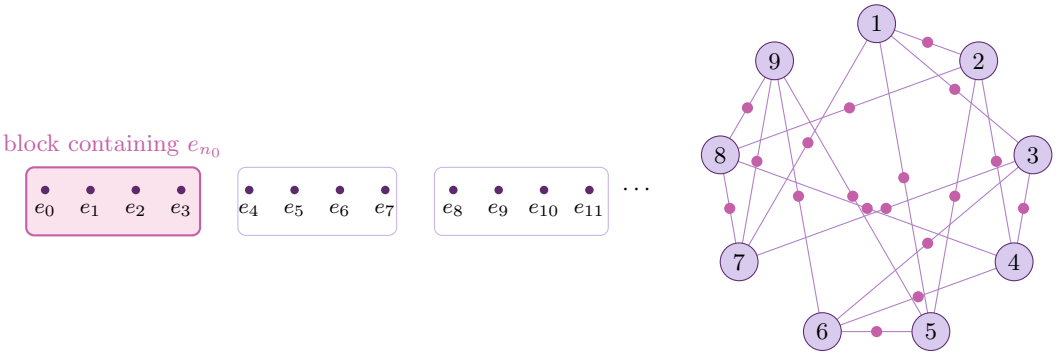


FIGURE 12. The finite obstruction at a leaf. Left: the basis vectors e_n grouped in blocks of four, with the block that contains e_{n_0} highlighted (here $n_0 = 2$, as in Figure 11). Right: the nine bases of Table 1 (numbered circles) and their eighteen rays (dots), each ray drawn on the segment that joins the two bases containing it. Since the block has measure 1, a two-valued measure would give the value 1 to exactly one ray in each basis; counting these rays basis by basis would count each of them twice and give an even number, although there are nine bases.

By Proposition 10.6(3), Josh gives countable answers in some plays of every winning strategy of Olivia in $\mathrm{G}(T^{01})$, and by Corollary 10.5(5), the plays of such a strategy have bounded length once Josh's countable answers are fixed. We do not know whether the length can be bounded independently of those answers.

Question 10.7. Does Olivia have a winning strategy in $\mathrm{G}(T^{01})$ whose tree has finite rank, that is, such that every play in which she follows it ends within a fixed number of rounds?

Remark 10.8 (Standard models). The game $\mathrm{G}(T)$ concerns all models of T, standard or not, whereas Theorem B concerns standard models only, and standardness is not expressed by axioms. For a Borel set B that meets $[0,1]$, let T_B be the theory T together with the axiom $\sup_a \mathbf{1}_B(\mu(a)) \leq 0$, the normal form of the axiom of Theorem B; here $\mathbf{1}_B(\mu(a))$ is the formula of Lemma 5.1, whose value at a is 1 when $\mu(a) \in B$ and 0 otherwise. Olivia has a winning strategy in $\mathrm{G}(T_B)$ exactly when T_B has no models at all, standard or not. This is the case when B contains 0 or 1, by Lemma 9.1(3) and (M2), and when B contains $(0,1)$, by Corollary 9.2; for the remaining sets B, whether T_B has a model is open.

Historical note. The Banach–Mazur game began as Problem 43 of the Scottish Book, posed by Mazur; on 4 August 1935 Banach wrote in the book that Mazur's conjecture is true, but his proof was never published, and Oxtoby published one in 1957 [Mau81, Oxt57]. Gale and Stewart proved in 1953 that open and closed games are determined [GS53]; Dénes Kőnig had published his lemma on finitely branching trees in 1927 [Kőn27]. The witnesses of the game descend from the new constants of Henkin's completeness proof of 1949 [Hen49]. For Väänänen there are essentially three kinds of games in logic, the semantic game, the model existence game and the Ehrenfeucht–Fraïssé game, and his book of 2011 is built on them [Vää11].

11. Remarks

In these remarks we explain why the vocabulary links vectors to elements of the Solèr sort through projections, and why standardness cannot be dropped from Theorems A and B, nor axiom (K4) from Theorem C. We then describe the changes needed for real and quaternionic scalars, and we record the logical form of the axioms and the redundancies among them.

Remark 11.1 (Projections rather than membership). The vocabulary links a vector and an element of the Solèr sort through the projection π rather than through a membership relation: a vector x belongs to a exactly when $\|\frac{1}{2}x - \frac{1}{2}\pi(a,x)\| = 0$. The choice matters for joins, because in real-valued logic an existential quantifier becomes an infimum. In a standard model whose Hilbert sort is the closed unit ball of a Hilbert space H, the condition "$x = y+z$ for some member y of a and some member z of b", read in this way, holds exactly at the sums $y+z$ of such members; these sums form a closed set, because bounded sequences in the closed subspaces S_a and S_b spanned by the members of a and of b have weakly convergent subsequences. When S_a+S_b is not closed, this set therefore misses the vectors of the unit ball that lie in the join $\overline{S_a+S_b}$ of S_a and S_b in $L(H)$ but not in S_a+S_b. Axiom (Sub3) reaches the join through the projections instead, by von Neumann's alternating projection theorem. The functions at and rep provide the bijection between the rays of H, which correspond to the unit vectors up to a phase, and the atoms of the Solèr sort (see Proposition 7.4). Completeness of the Solèr sort is not an axiom: in standard models it is a consequence of the axioms of T_1 (see Theorem 8.3).

Remark 11.2 (Nonstandard models). Standardness cannot be dropped from Theorem B or from Theorem A(1): by the downward Löwenheim–Skolem theorem, the theory T has countable models, in which the range of μ is countable and the Solèr sort, being countable, is not of the form $L(H)$.

In a nonstandard model of T_0, even the conclusion of Gleason's theorem for T_0 (see Theorem 7.7) can fail. Let E_0 be the space of finitely supported complex sequences with the inner product of ℓ^2, and let Λ_0 be the lattice of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0 (see Example 4.3). Take the closed unit ball of E_0 as Hilbert sort and Λ_0 as Solèr sort, with the orthogonal projections onto its elements, $\mathrm{at}(x) = \mathbb{C}x$, a

choice of rep, and the standard unit vectors as (e_n). The resulting structure is a model of T_0 that is not standard. Every maximal orthonormal family of E_0 is countable and has join 1 in Λ_0; a σ-additive state on Λ_0 would therefore yield a frame function on E_0, a function on the unit vectors whose sum over every maximal orthonormal family is 1 and whose restriction to the unit vectors of every finite-dimensional subspace is a frame function of that subspace. By a theorem of Dvurečenskij [Dvu95, Theorem 3.1] (compare Hamhalter and Pták [HP87]), an inner product space that carries such a function is complete, and E_0 is not. Hence Λ_0 has no σ-additive states. On the other hand, every density operator ρ on ℓ^2 defines a map $a \mapsto \mathrm{Tr}(\rho P_a)$ on Λ_0, with P_a the projection of ℓ^2 onto the closure of a, and the conclusion of Theorem 7.7, which identifies the σ-additive states with these maps, therefore fails for this nonstandard model.

The axioms of Theorem B exclude, one Borel set at a time, every range other than $[0,1]$ in standard models; no single sentence, however, says that the range of μ is disconnected. Indeed, suppose that a sentence ψ of real-valued $L_{\omega_1,\omega}$ in the vocabulary of T took the value 0 exactly in the models of T in which the range of μ is disconnected. In a standard model $\mathcal{M}$ of T, the range of μ is $[0,1]$ by Theorem B, and so the value of ψ in $\mathcal{M}$ is not 0. By the downward Löwenheim–Skolem theorem, the model $\mathcal{M}$ has a countable elementary substructure $\mathcal{N}$ for a countable fragment containing T and ψ; the value of ψ in $\mathcal{N}$ is also not 0, yet the range of μ in $\mathcal{N}$ is countable and contains $\mu(0) = 0$ and $\mu(1) = 1$, and it is therefore disconnected, a contradiction.

We leave open whether, in a nonstandard model of T, the measure μ can fail to be of the form $a \mapsto \mathrm{Tr}(\rho \bar{P}_a)$ for a density operator ρ on the Hilbert space $\bar{E}$ of Lemma 6.3.

Axiom (K4) is essential to Theorem C: the structure described in the proof of Proposition 10.6(2), with S empty, satisfies every axiom of T except (K4), and its measure takes only the values 0 and 1.

Remark 11.3 (Scalars). The vocabulary fixes the complex numbers as scalars. For real Hilbert spaces, replace $\mathbb{Q}(i)$ by $\mathbb{Q}$ and omit the predicate $\mathrm{Im}\langle x \mid y\rangle$; for quaternionic spaces, use the quaternions with rational coordinates, acting on the left, and four predicates for the real components of the form. In the reading of (At4), "complex number of modulus one" then becomes "real number of modulus one" or "quaternion of modulus one". Each choice yields the analogue of Theorem 8.3 for the corresponding case of Theorem 3.6, and Theorem C holds in all three cases, since the vectors of Table 1 have real coordinates.

Remark 11.4 (The logical form of the axioms). The axioms (H1), (OL), (P1)–(P3), (Sub0), (Sub4), (At1), (At2), (K1), (K2), (K4), (Sp1), (M1) and (M2) are universal conditions built from finitary formulas. Axioms (H2), (Sub2), (Sub3), (K3), (Sp2), (M3) and the formula θ use countable suprema or infima of uniformly bounded families. Axioms (P4), (Sub1), (At4), (At5), (Sp2), (Sp3) and (M3) use the device (2), which needs only uniform bounds. The functions at and span take values in the Solèr sort, where d is discrete, and they are not continuous from the norm to d; the logic of Section 5 permits this. Axiom (At3) uses the formula $\chi(a)$, which contains a quantifier over the Solèr sort.

Remark 11.5 (Redundancy). In standard models, several axioms follow from others: (Sub1) from (Sub2) and (P4); (Sub3) from (Sub2), (Sub4), (OL) and (P4); (At3) from (At1), (At2) and Proposition 7.3; and (At4) from (At1), (P4) and Proposition 7.3. These axioms are kept nonetheless, so that each property has an axiom of its own.

Historical note. Löwenheim proved in 1915 that a first-order sentence with a model has a countable model [Lö15]. Skolem extended the result to sets of sentences, and in 1922 he observed that axiomatic set theory, in which uncountable sets exist, therefore has countable models [Sko70]. The countable models of Remark 11.2 are of the same kind: they satisfy every

axiom of a theory whose standard models are uncountable. That an incomplete inner product space can be recognized by the absence of states goes back to Hamhalter and Pták, who proved in 1987 that a separable inner product space is complete exactly when the lattice of its orthogonally closed subspaces admits a σ-additive state [HP87]. The theorem of Dvurečenskij used in Remark 11.2 belongs to the same line of work [Dvu95].

Appendix A. The model existence game and Keisler's forcing

Keisler [Kei73] derives the model existence theorem and the omitting types theorem from a single result on forcing, which he describes as a reformulation, in terms of ordinary models, of the Boolean-valued models of Rasiowa and Sikorski [RS63]. In this appendix we compare his treatment with the game of Section 10; the comparison is clearest for the sentences that Keisler calls $\forall\exists$, whose challenges in the game reproduce one step of his forcing.

Keisler's forcing. Let C be a countable set of new constants. A *forcing property* is a partial order $\langle P, \leq\rangle$ with a least element, whose elements are the *conditions*, together with a map f that assigns to each condition p a set $f(p)$ of atomic sentences in the constants of C, increasing in p, subject to closure conditions on equality; among them, for every closed term τ and every p there are $c \in C$ and $q \geq p$ with $(c = \tau) \in f(q)$. The forcing relation is defined by recursion: $p \Vdash \varphi$ for atomic φ if $\varphi \in f(p)$; $p \Vdash \neg\varphi$ if no $q \geq p$ forces φ; $p \Vdash \bigvee \Phi$ if $p \Vdash \varphi$ for some $\varphi \in \Phi$; and $p \Vdash \exists x \varphi(x)$ if $p \Vdash \varphi(c)$ for some $c \in C$. Then $p \Vdash \forall x \varphi(x)$ if and only if, for every $c \in C$ and every $q \geq p$, some $r \geq q$ forces $\varphi(c)$, and similarly for $\bigwedge \Phi$ [Kei73, Lemma 1.3]. A *generic set* is a filter of conditions that contains, for each sentence of a fixed countable fragment, a condition that forces it or its negation. By the Generic Model Theorem, every condition belongs to a generic set, and every generic set G generates a model, whose elements are classes of constants and in which a sentence holds if and only if some condition in G forces it [Kei73, Lemmas 1.4 and 1.5].

The dictionary. The game distributes the steps of this construction between the two players. Positions, ordered by inclusion, play the role of conditions, and witnesses play the role of the constants of C; the atomic information $f(p)$ corresponds to the records of a position and to the intervals that it commits for basic sentences, since a real-valued atomic sentence is confined to an interval rather than declared true or false. The requirement that every closed term be equated with a constant is the pair of rules (Tot) and (Tm). The existential clauses of the forcing relation, for $\exists$ and $\bigvee$, are Josh's choices: a witness under $(\sup \geq)$ and $(\inf \leq)$, an index under $(\sup_n \geq)$ and $(\inf_n \leq)$. The universal clauses, for $\forall$ and $\bigwedge$, have the shape of a round: Olivia chooses the constant or the conjunct, under $(\sup \leq)$, $(\inf \geq)$, $(\sup_n \leq)$ and $(\inf_n \geq)$, and the round at which she does so, which plays the role of the arbitrary extension $q \geq p$; Josh's answer, which these rules determine, plays the role of the extension $r \geq q$. A fair play in which Josh never loses corresponds to a generic set, the proof that (2) implies (1) in Theorem 10.3 corresponds to Keisler's construction of the generic model, with the passage to a reduced structure in place of his identification of constants c and d when $c = d$ is forced, and the Baire category argument of Remark 10.4 corresponds to the theorem of Rasiowa and Sikorski.

$\forall\exists$ sentences. Keisler calls a sentence $\forall\exists$ when it has the form

$$\forall x_1 \cdots \forall x_m \bigvee_{n \in \mathbb{N}} \exists y_1 \cdots \exists y_{l_n} \left(\varphi_{n1} \wedge \cdots \wedge \varphi_{nj_n}\right),$$

with each φ_{ni} atomic or negated atomic. For a class $\mathcal{M}$ of models, a *finite piece* of $\mathcal{M}$ is a finite set of atomic and negated atomic sentences in the constants of C that holds in some model of $\mathcal{M}$ under some assignment of the constants; the finite pieces, ordered by inclusion, form a forcing property. Keisler's Theorem 2.1 says that the $\forall\exists$ sentence above holds in every generic model for this forcing property if and only if, for every finite piece p and every tuple $\bar{c}$ of constants, there are n and constants $\bar{d}$ such that $p \cup \{\varphi_{n1}(\bar{c},\bar{d}),\dots,\varphi_{nj_n}(\bar{c},\bar{d})\}$ is again a finite piece [Kei73, Theorem 2.1 and its proof]. His Basic Omitting Types Theorem follows: if $\mathcal{M}$ is the class of models of a set of $\forall\exists$ sentences and each sentence of a countable family of $\forall\exists$ sentences satisfies this condition, then $\mathcal{M}$ has a countable model of the whole family.

The real-valued counterpart of a $\forall\exists$ sentence is an axiom

$$\sup_{\bar{x}} \inf_{n} \inf_{\bar{y}} \theta_n(\bar{x},\bar{y}) \le 0,$$

with quantifier-free formulas θ_n and a common bound, which says that for every $\bar{x}$ and every $\varepsilon > 0$ there are n and $\bar{y}$ with $\theta_n(\bar{x},\bar{y}) \le \varepsilon$. In the game, a challenge (Ax) to this axiom opens a block of moves with the shape of Keisler's condition.

(1) *The universal quantifiers.* Under $(\sup \le)$, Olivia names witnesses $\bar{c}$, and Josh commits $\inf_n \inf_{\bar{y}} \theta_n(\bar{c},\bar{y})$ to an interval with right endpoint 0.
(2) *The countable disjunction.* Under $(\inf_n \le)$, Olivia names a precision, and Josh names an index n.
(3) *The existential quantifiers.* Under $(\inf \le)$, Olivia names precisions, and Josh names witnesses $\bar{d}$, which already occur in the position or are fresh, and commits $\theta_n(\bar{c},\bar{d})$ to an interval whose right endpoint is close to 0.
(4) *The quantifier-free matrix.* Under (Cn) and (Tm), Olivia decomposes $\theta_n(\bar{c},\bar{d})$, and Josh commits the basic sentences in it to dyadic cells.

Keisler's condition says that, at every condition and for every $\bar{c}$ that Olivia may name, Josh has answers n and $\bar{d}$ that keep the atomic information among the conditions; the omitting types theorem says that Josh can then answer such blocks for all the sentences of a countable family without leaving the conditions, and the generic model satisfies them all. The only infinite choice in the block is the index n in step (2), so that, by Corollary 10.5(3), a winning strategy of Olivia branches infinitely only there. When the disjunction is finite, the sentence is first-order, the branching is finite, and Corollary 10.5(4) gives compactness. We do not attempt here a real-valued version of Keisler's Theorem 2.1 stated in terms of the game.

The axiom $\sup_a \min(\mu(a), 1-\mu(a)) \le 0$ of the theory T^{01} of Proposition 10.6 is universal: a challenge to it involves no choice by Josh except the cells under (Cn). By Proposition 10.6(2) and (3), every finite subset of T^{01} has a model, and every winning strategy of Olivia passes through a countable answer of Josh, that is, through step (2) of a block of the kind just described. In the strategy described in Section 10.4, the decisive block comes from σ-additivity, axiom (M3), at the sequence ε: the commitment that $\sup_N \min\bigl(2, \sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))\bigr)$ is close to 1 is an instance of the $\forall\exists$ pattern, with the universal quantifier already instantiated at ε and an empty existential part: it asks for some N whose partial sum is close to 1, and Josh names N. The finite Kochen–Specker configuration enters only after this choice, at the leaves.

The difference. In Keisler's applications the conditions are satisfiable by design, as the finite pieces of a class of models are, so every condition belongs to a generic set, and the work is to show that the generic models have the required properties; in his proof of an end extension theorem of Barwise, "the only difficulty" is to show that the set of conditions is nonempty [Kei73, p. 130]. The game starts from arbitrary finite positions, constrained

only by feasibility. The positions from which Josh has a winning strategy form a family of feasible positions in which every available challenge has a legal answer that stays in the family, and the theory has a model exactly when this family contains the empty position. Olivia's winning strategy certifies that it does not, by the well-founded tree of finite obstructions of Corollary 10.5. Model-theoretic forcing in an infinitary continuous logic was developed by Ben Yaacov and the second author [BYI09]; the game defines no forcing relation for commitments.

Historical note. Robinson introduced forcing into model theory [Rob71], adapting Cohen's method in set theory [Coh63], and Barwise extended Robinson's forcing to countable fragments of $L_{\omega_1,\omega}$ and used it to give a new proof of the omitting types theorem, in notes that Keisler cites [Kei73]. The omitting types theorem itself goes back to Henkin [Hen54] and Orey [Ore56]. Keisler's article of 1973 derives the omitting types theorem and Cohen's independence results from one Generic Model Theorem, which he attributes essentially to Rasiowa and Sikorski [RS63].

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Department of Mathematics, UT San Antonio, One UTSA Circle, San Antonio, TX 78249, USA

Email address: olivia.aubone@my.utsa.edu

Department of Mathematics, UT San Antonio, One UTSA Circle, San Antonio, TX 78249, USA

Email address: jose.iovino@utsa.edu